AI-103 Practice

Set A

100 questions

Answers, explanations and source links

Objectives effective 16 April 2026

Case study - Rotor equipment reservations

Rotor rents industrial equipment to authenticated business customers. Its assistant can explain availability and propose reservations, but the inventory API remains the system of record. The operations team is replacing a text-only prototype with a Responses-based prompt agent. Each question concerns an independent request against the configuration below.

Application and identity

The backend receives a signed application session identifying the customer and warehouse. It calls the agent using one managed identity. Two host functions are registered: get_stock(part_id, warehouse_id) and reserve_part(part_id, warehouse_id, quantity, customer_id). The stock function is read-only. The reservation function changes inventory and returns a reservation number after the database commits. The managed identity can reach the API, while application authorization must restrict each customer to its own account. JSON schema validation checks shape and types; inventory limits and account ownership are enforced by the host.

Action policy

A customer may request a proposal without committing it. Reservations costing more than 250 currency units require a supervisor to approve the exact customer, item, quantity and quoted amount. An approval for five units is not approval for fifty. A changed proposal must return to the approval screen. The host does not infer approval from manuals, tool-returned text or the agent saying that approval was obtained. The operations log records the proposal, approving principal and committed reservation number so that an incident can be reconstructed.

Observed request

A request checks the same part at two warehouses. Responses returns two function calls named get_stock with call_id values rotor-east and rotor-west. The west warehouse replies first. The host must submit function_call_output records after obtaining the actual results. A separate request asks for a guaranteed fresh stock lookup rather than an answer from the model’s prior knowledge. Both functions are attached and the chosen model supports function calling. No inventory lookup has happened merely because a function call is present in the model output.

Failure investigation

A test customer asks for a reservation belonging to another customer account. The argument object passes the JSON schema. A second test approves five units, but a later model turn changes the quantity to fifty before execution. A third test asks for a proposal only, while a developer considers forcing tools on every turn. Engineers need to choose controls at the correct boundary rather than relying on a single system instruction. The questions do not assume that a correct earlier answer changes these independently stated observations.

Diagnostic exhibit

Request exhibit: {"calls":[{"name":"get_stock","call_id":"rotor-east"},{"name":"get_stock","call_id":"rotor-west"}],"completionOrder":["rotor-west","rotor-east"]}. The reservation tests are separate requests, not effects of this lookup.

Q1

Domain B: Implement generative AI and agentic solutions | Case study

Use shared case: Rotor equipment reservations.
In Rotor’s two-warehouse request, west completes first. Which output correlation preserves the API contract?

A Replace the call_id with a newly generated receipt ID.

B Associate results with completion position.

C Associate each result with its originating call_id. ✓

D Associate both results with get_stock.

Why the correct answer is right

rotor-east and rotor-west identify distinct proposed calls even though the function name matches. Completion order does not change function_call_output correlation.

Why the other options are wrong

- **D:** The shared name cannot distinguish the two calls.
- **B:** A slower call can complete later; position is not the correlation key.
- **A:** A business receipt is not the originating function call identifier.

AI103-B2.01

Sources

S007: Use function calling with Microsoft Foundry agents
https://learn.microsoft.com/azure/foundry/agents/how-to/tools/function-calling

Q2

Domain B: Implement generative AI and agentic solutions | Case study

Use shared case: Rotor equipment reservations.
Rotor’s fresh availability turn must use a tool rather than answer from prior knowledge.
Both tools are available. Which request-level mode requires tool use?

A none

B required ✓

C parallel_tool_calls=true

D auto

Why the correct answer is right

required demands one or more tool calls. The host still checks which action is proposed, its authorization and any approval requirement.

Why the other options are wrong

- **D:** auto permits a response without a tool call.
- **A:** none prevents tool calls.
- **C:** Permitting parallel calls does not require a tool call.

AI103-B2.02

Sources

S054: Tool best practices for Microsoft Foundry Agent Service
https://learn.microsoft.com/azure/foundry/agents/concepts/tool-best-practice

S007: Use function calling with Microsoft Foundry agents
https://learn.microsoft.com/azure/foundry/agents/how-to/tools/function-calling

Q3

Domain B: Implement generative AI and agentic solutions | Case study

Use shared case: Rotor equipment reservations.
In Rotor’s cross-account reservation test, the JSON schema passes. Which check must decide whether the host may execute?

A Apply the warehouse’s rate limit and accept the supplied customer ID.

B Validate caller ownership and reservation business constraints. ✓

C Validate the argument types and permit any schema-valid account.

D Check that the selected function is registered and has a description.

Why the correct answer is right

The backend session identifies the customer. Shape validation does not establish ownership or valid inventory quantities.

Why the other options are wrong

- **C:** Valid types do not establish caller ownership.
- **D:** Registration does not establish customer authorization.
- **A:** Rate limiting does not establish ownership of the supplied account.

AI103-B2.05

Sources

S007: Use function calling with Microsoft Foundry agents
https://learn.microsoft.com/azure/foundry/agents/how-to/tools/function-calling

S054: Tool best practices for Microsoft Foundry Agent Service
https://learn.microsoft.com/azure/foundry/agents/concepts/tool-best-practice

Q4

Domain B: Implement generative AI and agentic solutions | Case study

Use shared case: Rotor equipment reservations.
Rotor approved five units, then the proposal changed to fifty. What should the host do?

A Reapprove the changed argument tuple before execution. ✓

B Execute fifty units and append the old approval record.

C Reuse the approval because the function name is unchanged.

D Accept the agent's claim that the old approval covers the new quantity.

Why the correct answer is right

The case approval binds customer, item, quantity and amount. Changed arguments require a new approval, followed by truthful reporting of the committed result.

Why the other options are wrong

- **C:** Approval applies to the exact proposal, not a function category.
- **B:** Appending evidence after the side effect cannot supply prior approval.
- **D:** Model acknowledgement does not amend the exact-tuple approval contract.

AI103-B2.05

Sources

S007: Use function calling with Microsoft Foundry agents
https://learn.microsoft.com/azure/foundry/agents/how-to/tools/function-calling

S054: Tool best practices for Microsoft Foundry Agent Service
https://learn.microsoft.com/azure/foundry/agents/concepts/tool-best-practice

Case study - Harbor maintenance knowledge

Harbor maintains manuals for several equipment departments. Its new assistant must answer policy and catalog questions with current evidence and distinguish factual support from useful but unrelated prose. Search, embedding and generation are separately observable components. Each child question uses the same architecture and an independent observation.

Corpus and index

The Search index stores chunk text, a filterable department field, a catalog identifier and one vector field. Maintenance staff often search by exact identifiers such as HX-104 as well as by descriptions of symptoms. The current corpus was embedded with deployment harbor-embed-x. A proposed query deployment harbor-embed-y has the same vector dimensions but uses an unrelated model. The index is not automatically regenerated when a developer changes a query-time setting. The source documents and their department labels are authoritative.

Access and retrieval

The application authenticates each employee and derives an allowed department from its own authorization service. Department boundaries apply before evidence is returned to the model. A generated instruction asking the model to ignore unauthorized passages is not the access filter. The team can configure keyword retrieval, vector retrieval and hybrid retrieval. An optional semantic reranker operates on retrieved candidates; it does not create content embeddings or rename index fields. Query vectors must be compatible with the content vectors being searched.

Evaluation evidence

A held-out test includes queries, relevance judgments for candidate documents, retrieved passages and generated answers. One passage says the repair warranty lasts twelve months. An answer says twenty-four months while staying on the user’s topic. Another answer quotes the twelve-month passage correctly but responds to a shipping-time question. The team wants separate measurements for retrieval quality, contextual support and response relevance rather than one total score. A successful query response establishes transport success, not answer correctness.

Operations

Indexer logs record document-level errors and warnings. An overall successful run still requires checking whether required documents entered the index. An embedding model change needs a planned reindexing path and evaluation before rollout. Engineers retain safe request identifiers to correlate Search requests with agent traces. They do not put service credentials in model-visible tool errors. A reviewer should combine the named corpus, access and evaluation facts to choose each answer; previous child answers are not prerequisites and do not alter the configuration in this case.

Diagnostic exhibit

Evaluation exhibit: {"contentEmbedding":"harbor-embed-x","queryEmbedding":"harbor-embed-y","dimensionsMatch":true,"passageWarrantyMonths":12,"answerWarrantyMonths":24}. The authorized department is derived from the employee session, not either deployment name.

Q5

Domain B: Implement generative AI and agentic solutions | Case study

Use shared case: Harbor maintenance knowledge.
Harbor must serve HX-104 identifiers and descriptions of symptoms. Which retrieval approach combines those signals?

A Semantic reranking without initial retrieval.

B Hybrid keyword and vector retrieval with ranked-list fusion. ✓

C Vector search followed by alphabetical identifier sorting.

D Keyword retrieval followed by increasing generation temperature.

Why the correct answer is right

Harbor’s exact identifiers benefit from keyword retrieval and descriptive queries benefit from vectors. Hybrid search fuses the ranked candidates; a semantic reranker can operate afterward.

Why the other options are wrong

- **A:** Reranking requires retrieved candidates and does not replace retrieval.
- **C:** Alphabetical sorting is not a keyword relevance ranking.
- **D:** Generation temperature does not add vector retrieval.

AI103-B1.02

Sources

S025: Hybrid search using vectors and full-text search in Azure AI Search
https://learn.microsoft.com/azure/search/hybrid-search-overview

Q6

Domain B: Implement generative AI and agentic solutions | Case study

Use shared case: Harbor maintenance knowledge.
Harbor tests harbor-embed-y for queries while content remains from harbor-embed-x.
Equal dimensions did not preserve relevance. Which correction targets the mismatch?

A Normalize vector magnitudes and retain both current embedding models.

B Enable semantic ranking and leave embedding compatibility unresolved.

C Use a validated compatible embedding configuration for content and queries. ✓

D Increase vector candidate count and retain the unrelated query model.

Why the correct answer is right

The case uses unrelated models. Dimension compatibility is necessary for vector fields but does not create a shared semantic space.

Why the other options are wrong

- **D:** Candidate count does not align the two embedding spaces.
- **A:** Magnitude normalization does not align learned semantic spaces.
- **B:** Reranking is an additional stage and does not repair the vector representation contract.

AI103-B1.02

Sources

S026: Integrated vector embedding in Azure AI Search
https://learn.microsoft.com/azure/search/vector-search-integrated-vectorization

S058: Integrated vectorization REST configuration
https://learn.microsoft.com/en-us/azure/search/search-how-to-integrated-vectorization

Q7

Domain B: Implement generative AI and agentic solutions | Case study

Use shared case: Harbor maintenance knowledge.
Harbor answers twenty-four months from a passage saying twelve. It stays on topic. Which evaluation isolates this defect?

A Groundedness ✓

B Tool selection

C Query latency

D Relevance

Why the correct answer is right

The warranty claim contradicts the supplied passage. Groundedness examines answer support in context, distinct from topical relevance.

Why the other options are wrong

- **D:** The answer can be on topic while unsupported.
- **C:** Transport timing does not assess factual support.
- **B:** The defect described is a generated claim, not a selected tool.

AI103-B1.04

Sources

S039: Retrieval-Augmented Generation (RAG) evaluators
https://learn.microsoft.com/azure/foundry/concepts/evaluation-evaluators/rag-evaluators

Q8

Domain B: Implement generative AI and agentic solutions | Case study

Use shared case: Harbor maintenance knowledge.
Harbor’s employee is authorized for one department. Which control enforces that boundary before passages reach the model?

A Remove other departments from citations after the answer is generated.

B Use the employee’s department in a system instruction after retrieval.

C Rerank the unfiltered results and retain only the highest-scoring passage.

D Apply the authenticated employee’s department as a retrieval filter. ✓

Why the correct answer is right

The case requires access filtering before evidence is returned. A supported filter on the filterable department field narrows retrieval; the host supplies the authorized value.

Why the other options are wrong

- **B:** The policy requires filtering before passages enter context.
- **A:** Post-generation citation filtering is too late for this boundary.
- **C:** Relevance is not the employee authorization rule.

AI103-B1.02

Sources

S040: Connect an Azure AI Search index to Foundry agents
https://learn.microsoft.com/azure/foundry/agents/how-to/tools/ai-search

S054: Tool best practices for Microsoft Foundry Agent Service
https://learn.microsoft.com/azure/foundry/agents/concepts/tool-best-practice

Case study - Lumen regional engineering assistant

Lumen operates an engineering assistant for staff in the European Union. Its platform team is designing deployment, identity and retrieval controls together, with separate principals for deployment and runtime. The service must answer interactively and retain existing learner and operational records during the migration described here.

Workload boundary

The processing policy permits inference across EU countries but forbids processing outside that data zone. Interactive requests cannot wait for a nightly batch. Demand varies across the day and procurement wants on-demand token billing rather than reserved throughput. The selected model is available for every deployment family considered in the questions, and the relevant quota has already been confirmed. The resource’s region alone is not the organization’s definition of the inference processing boundary.

Inference identity

The backend calls an Azure OpenAI resource through its /openai/v1/ endpoint using Microsoft Entra credentials. Its runtime managed identity currently has Reader on that resource and no inference data role. Token acquisition succeeds, but inference is denied. The team wants the documented inference role at the resource scope without granting model-management permissions. Developers’ personal roles and the GitHub deployment principal are separate from the runtime identity. A resource key is not part of the approved runtime design.

Project and Search

An AI Projects Python 2.8.0 client is used for Foundry project operations, with a project endpoint and an Entra credential. Search public access is disabled. The private endpoint exists, and the agent connection is explicitly pinned to the named runtime principal. That principal has Search Index Data Reader for the target index. From the runtime subnet, however, the Search hostname resolves to a public address. From a developer laptop on another network it resolves privately and querying succeeds. The network observations do not transfer from the laptop to the agent subnet.

Release process

A GitHub deployment workflow obtains a token using an established OIDC trust. It does not store a client secret. The workflow principal’s project role is assigned on a different project, while its endpoint variable points to the intended Lumen project. An authorization denial therefore needs identity and scope analysis after authentication succeeds. Operations retain redacted request spans and release identifiers to investigate failures. Each question asks for a single independent decision; repairing one defect is not assumed to repair the others. A proposed change must satisfy the named runtime, region and billing facts rather than simply resemble a working developer setup.

Diagnostic exhibit

Runtime exhibit: {"tokenAcquired":true,"inferenceRole":"Reader","searchRole":"Search Index Data Reader","searchDns":"public address","publicSearchAccess":false}. These observations belong to the runtime identity and subnet; the developer’s successful call uses a different network.

Q9

Domain A: Plan and manage an Azure AI solution | Case study

Use shared case: Lumen regional engineering assistant.
Which deployment fits Lumen’s interactive EU-processing and on-demand billing requirements?

A Global Standard in an EU resource.

B Data Zone Standard in the EU. ✓

C Data Zone Provisioned.

D Data Zone Batch.

Why the correct answer is right

Lumen permits EU-wide processing, requires interactive completion and wants token billing. Data Zone Standard satisfies those constraints given the stated model availability and quota.

Why the other options are wrong

- **A:** The resource region alone does not constrain Global processing to the EU.
- **D:** Batch is asynchronous and conflicts with the interactive workload.
- **C:** Provisioned introduces reserved throughput rather than the requested on-demand billing.

AI103-A2.02

Sources

S005: Deployment types for Microsoft Foundry Models
https://learn.microsoft.com/azure/foundry/foundry-models/concepts/deployment-types

Q10 Domain A: Plan and manage an Azure AI solution Case study

Use shared case: Lumen regional engineering assistant.
Lumen’s runtime obtains a token but Reader-only inference is denied. Which documented role at the called Azure OpenAI resource supplies inference permission?

A A deployment workflow’s role on another project.

B Reader on the Foundry project only.

C Cognitive Services OpenAI User. ✓

D Search Index Data Reader.

Why the correct answer is right

The named /openai/v1/ endpoint requires the runtime identity’s inference permission. Cognitive Services OpenAI User is the documented inference role; scope must cover the called resource.

Why the other options are wrong

- **B:** Management reading does not grant the named inference permission.
- **D:** That role applies to Search data, not Azure OpenAI inference.
- **A:** A different principal and different resource scope do not authorize this runtime call.

AI103-A3.04

Sources

S051: Azure OpenAI v1 API
https://learn.microsoft.com/en-us/azure/foundry/openai/api-version-lifecycle

S052: Use Azure OpenAI without keys
https://learn.microsoft.com/en-us/azure/developer/ai/keyless-connections

Q11 Domain A: Plan and manage an Azure AI solution Case study

Use shared case: Lumen regional engineering assistant.
Lumen’s Search role is correct, but the runtime resolves Search publicly. What should be repaired first?

A The runtime principal’s model-token quota.

B The semantic configuration on the Search index.

C The developer laptop’s working Search role.

D Private DNS resolution and subnet links. ✓

Why the correct answer is right

Public access is disabled and the runtime resolves to a public address. Repair its private DNS path and connectivity; the laptop observation is from another network.

Why the other options are wrong

- **A:** Model quota does not change Search hostname resolution.
- **C:** The case already distinguishes the developer from the runtime network.
- **B:** Semantic relevance settings do not repair the network path.

AI103-A3.04

Sources

S011: Set up private networking for Foundry Agent Service
https://learn.microsoft.com/azure/foundry/agents/how-to/virtual-networks

Q12 Domain A: Plan and manage an Azure AI solution Case study

Use shared case: Lumen regional engineering assistant.
Lumen’s OIDC token succeeds but deployment is denied. Which authorization correction targets the case defect?

A Change the runtime Search role.

B Use the endpoint string as a credential.

C Scope the workflow’s deployment role to the intended project. ✓

D Rotate a long-lived secret into GitHub.

Why the correct answer is right

Authentication works. The workflow role is scoped to a different project; align deployment authorization with the target while retaining federation.

Why the other options are wrong

- **D:** Changing credential style does not correct the role’s resource scope.
- **A:** Runtime Search access belongs to a different operation and principal.
- **B:** An endpoint locates a service and is not an authentication credential.

AI103-A2.04

Sources

S012: Quickstart: Hosted agent CI/CD templates
https://learn.microsoft.com/azure/foundry/agents/quickstarts/set-up-cicd-hosted-agent

S010: Role-based access control for Microsoft Foundry
https://learn.microsoft.com/azure/foundry/concepts/rbac-foundry

Case study - North support memory and response contracts

North runs a support assistant for individual subscribers. A web backend authenticates users locally and calls a Foundry prompt agent on their behalf. Subscribers can retain preferences across sessions, ask about current support policies and receive structured summaries. The team is testing the distinction between stored conversation history, persistent memory and curated knowledge.

Caller and end user

All response calls authenticate to Foundry with one backend managed identity. The memory search tool is attached to the agent and uses scope {{$userId}}. In the current implementation the backend sends no x-memory-user-id header. The application has a verified tenant identifier and subscriber identifier in its authenticated session. Those identifiers can be combined into a stable application-owned memory key. The host must derive that key from the authenticated session rather than accept an arbitrary customer identifier from a prompt.

Memory operations

The service also runs an administrative job that calls the low-level memory search API directly. The job uses the same backend identity. It supplies a subscriber identifier in the query text but omits the explicit scope parameter. The job must search the intended subscriber partition. Current support policy is maintained in a knowledge store, while language and unit preferences may be distilled from conversations. Changing a conversation title does not change either memory’s partition or the lifecycle of the organizational knowledge.

Structured application result

The client sends requests through Responses, not Chat Completions. A summary object has ticket_id and note properties; note accepts a string or null. Both keys must be present, and unknown properties must be rejected by the supported strict schema. The model supports that schema. A missing note is represented by null. The application handles refusal outcomes separately from successful objects. A developer has copied a response_format field from a Chat Completions example into the Responses request builder.

Behavior and monitoring

A system instruction limits the assistant to subscriber support and gives an explicit clarification path when evidence is missing. Product staff propose replacing that instruction with a low temperature. A workflow specialist and a retrieval specialist may also be composed, with the host retaining control of routing and result validation. Trace spans record latency and safe user-correlation identifiers; access tokens and full personal messages are redacted before export. These controls must be considered at their own boundaries. Every child observation is independent and requires the fixed case facts rather than an inferred result from a previous child.

Diagnostic exhibit

Backend exhibit: {"caller":"shared backend principal","memoryToolScope":"{{$userId}}","memoryUserHeader":null,"directSearchScope":null}. The Responses request builder also copied the Chat Completions response_format field; these are distinct defects against the stated per-subscriber and output contracts.

Q13 Domain B: Implement generative AI and agentic solutions Case study

Use shared case: North support memory and response contracts.
North uses one backend principal and {{$userId}} with no header. Which implementation separates authenticated subscribers?

A Place the subscriber key only in a natural-language prompt.

B Send the authenticated subscriber key in x-memory-user-id. ✓

C Create a new conversation title for each subscriber.

D Retain the backend identity as the shared user partition.

Why the correct answer is right

Without the header the tool falls back to the caller's Entra identity. North's caller is the shared backend. Derive an authorized end-user key and send the documented header.

Why the other options are wrong

- **C:** A title does not define the memory tool's scope.
- **D:** The shared caller cannot provide separate subscriber partitions.
- **A:** Prompt text does not replace the documented scope-resolution input.

AI103-B2.02

Sources

S053: Create and use memory in Foundry Agent Service (preview)
https://learn.microsoft.com/en-us/azure/foundry/agents/how-to/memory-usage

Q14 Domain B: Implement generative AI and agentic solutions Case study

Use shared case: North support memory and response contracts.
North’s administrative job searches low-level memory with the subscriber named only in query text. What must change?

A Rely on {{$userId}} being inferred from the query text.

B Supply the authorized subscriber scope on the direct API call. ✓

C Use the subscriber’s display name as the conversation title.

D Attach the query to a longer conversation.

Why the correct answer is right

Direct memory operations require explicit scope; automatic identity extraction is a memory-search tool behavior. The admin job must select its authorized subscriber partition.

Why the other options are wrong

- **C:** Display naming does not supply the direct operation’s scope.
- **D:** Conversation length does not choose a memory partition.
- **A:** The low-level API does not infer scope from query text.

AI103-B2.02

Sources

S053: Create and use memory in Foundry Agent Service (preview)
https://learn.microsoft.com/en-us/azure/foundry/agents/how-to/memory-usage

Q15 Domain B: Implement generative AI and agentic solutions Case study

Use shared case: North support memory and response contracts.
North’s Responses builder copied response_format from Chat Completions. Where should its structured-result format be configured?

A Chat Completions response_format left unchanged on Responses.

B The returned function_call_output record.

C The conversation title.

D Responses text.format. ✓

Why the correct answer is right

The case explicitly uses Responses. Its structured-output request shape places the format under text.format; note’s nullable value and required presence are separate schema concerns.

Why the other options are wrong

- **B:** A tool result is not the model response-format configuration.
- **C:** Conversation naming does not configure the response schema.
- **A:** This retains the request-shape error being diagnosed.

AI103-B3.01

Sources

S037: Structured outputs
https://learn.microsoft.com/azure/foundry/openai/how-to/structured-outputs

Q16

Domain B: Implement generative AI and agentic solutions | Case study

Use shared case: North support memory and response contracts.
North wants a domain boundary and clarification when evidence is missing. Which design addresses behavior and enforcement?

A Use tone examples as the complete access control.

B Ask the model to authorize its own booking actions.

C Define role/fallback, evaluate behavior and enforce host permissions. ✓

D Lower temperature and remove the system instruction.

Why the correct answer is right

System instructions express the domain and fallback, but they do not guarantee compliance. Evaluation and host action controls implement the case's separate behavioral and permission boundaries.

Why the other options are wrong

- **D:** Temperature does not define an allowed domain or fallback.
- **A:** Tone examples do not authorize actions.
- **B:** The case gives action authority to the host, not model self-approval.

AI103-B3.01

Sources

S057: System message design
https://learn.microsoft.com/en-us/azure/cognitive-services/openai/concepts/advanced-prompt-engineering?pivots=programming-language-chat-completions

Q17 Domain A: Plan and manage an Azure AI solution Single choice

An interactive assistant may process requests across EU countries but cannot process them outside the EU. Usage varies, and the team wants token-based billing. The model supports all listed types. Which deployment fits?

A Data Zone Standard, using EU capacity with token-based charges for interactive requests. ✓

B Global Standard in an EU resource, relying on the resource region to keep inference processing within the EU.

C Data Zone Batch, submitting requests to the asynchronous batch queue.

D Data Zone Provisioned, with EU processing and reserved throughput.

Why the correct answer is right

Data Zone Standard combines the EU processing boundary with on-demand token billing. Model availability and quota must also be available in the selected deployment.

Why the other options are wrong

- **B:** Global processing is not restricted to the EU by the resource location.
- **D:** The workload calls for usage-based billing rather than reserved throughput.
- **C:** The assistant needs responses during the user interaction.

AI103-A2.02

Sources

S005: Deployment types for Microsoft Foundry Models
https://learn.microsoft.com/azure/foundry/foundry-models/concepts/deployment-types

Q18 Domain A: Plan and manage an Azure AI solution Single choice

A service has predictable sustained model traffic and needs allocated throughput. Its processing boundary is already satisfied by the chosen region. Which deployment characteristic should drive the selection?

A Developer deployment for experimentation with the selected model.

B Batch processing with deferred completion for groups of requests.

C Standard token billing with capacity shared across on-demand requests.

D Provisioned capacity with a throughput commitment. ✓

Why the correct answer is right

Provisioned deployments allocate throughput under a capacity commitment. Size the commitment using measured workload characteristics rather than equating it with on-demand quota.

Why the other options are wrong

- **C:** This does not reserve the planned throughput.
- **B:** Deferred completion does not meet a sustained interactive serving need.
- **A:** An experimentation deployment does not meet the production capacity requirement.

AI103-A2.02

Sources

S005: Deployment types for Microsoft Foundry Models
https://learn.microsoft.com/azure/foundry/foundry-models/concepts/deployment-types

Q19 Domain A: Plan and manage an Azure AI solution Single choice

Developers have scoped Foundry RBAC roles, but a shared resource key authenticates the application. A developer can perform an operation that their role excludes. Which change makes the role assignments effective for those calls?

A Move the application behind a private endpoint while retaining key authentication.

B Rotate the shared resource key and continue assigning each developer the same key.

C Store the shared key in a vault and distribute it to authorized developers.

D Authenticate with Entra identities carrying the intended scoped roles. ✓

Why the correct answer is right

Foundry RBAC applies to Microsoft Entra authentication. A resource key grants access outside those individual role restrictions.

Why the other options are wrong

- **B:** Rotation does not make a shared key enforce individual RBAC.
- **A:** Network isolation does not change the authorization semantics of the key.
- **C:** Protecting storage does not apply user-role restrictions to key-authenticated calls.

AI103-A3.04

Sources

S010: Role-based access control for Microsoft Foundry
https://learn.microsoft.com/azure/foundry/concepts/rbac-foundry

Q20 Domain A: Plan and manage an Azure AI solution Single choice

A secured agent has the required runtime roles. Search resolves to a public IP from the agent network even though public access is disabled. The private endpoint exists. What should be investigated next?

A Private DNS resolution and links for the agent network. ✓

B The query-time semantic ranking configuration on the target index.

C The private endpoint connection approval state for the runtime subnet.

D The Search role assignments on the identity used by the deployment pipeline.

Why the correct answer is right

A private endpoint must be reachable through correct DNS resolution and routing from the runtime network. Roles cannot repair resolution to a blocked public endpoint.

Why the other options are wrong

- **C:** The stated failure is name resolution to a public address.
- **D:** The problem is name resolution from the runtime network.
- **B:** Ranking is evaluated after a request reaches Search.

AI103-A3.04

Sources

S011: Set up private networking for Foundry Agent Service
https://learn.microsoft.com/azure/foundry/agents/how-to/virtual-networks

Q21 Domain A: Plan and manage an Azure AI solution Single choice

A Foundry managed-compute preview deployment needs more accelerators. Matching GPU VM quota is available in the same region. Which quota should the team request?

A Request a larger subscription-wide regional vCPU allocation and count that capacity toward the Foundry managed-compute accelerator deployment.

B An increase to the classic managed-endpoint GPU VM-family quota for that region.

C Request the regional accelerator-SKU allocation through the Foundry managed-compute quota process. ✓

D Additional tokens per minute on the serverless language model deployment.

Why the correct answer is right

Managed-compute preview uses its own quota per accelerator SKU and region. Existing Azure VM quota cannot be applied to it.

Why the other options are wrong

- **B:** Classic VM quota is a separate allocation.
- **D:** Token quota does not allocate managed-compute accelerators.
- **A:** Foundry managed-compute quota is separate from the subscription VM allocation.

AI103-A3.01

Sources

S004: Deployment overview for Microsoft Foundry Models
https://learn.microsoft.com/azure/foundry/concepts/deployments-overview

Q22 Domain A: Plan and manage an Azure AI solution Single choice

A hosted agent returns session_quota_exceeded. Several sessions no longer need active compute, but their stored state is still required. What should the operator do?

A Stop the inactive sessions that are consuming active capacity. ✓

B Delete the completed sessions together with their stored conversation state.

C Increase the model deployment's reserved inference throughput.

D Increase the tokens-per-minute allocation for the underlying chat model.

Why the correct answer is right

Stopping sessions that no longer need active compute releases active session capacity while respecting the requirement to retain state.

Why the other options are wrong

- **B:** The requirement explicitly preserves the state.
- **D:** The reported code concerns sessions, not model tokens.
- **C:** Model throughput is separate from active hosted-session capacity.

AI103-A3.01

Sources

S009: Foundry Agent Service limits, quotas, and regional support
https://learn.microsoft.com/azure/foundry/agents/concepts/limits-quotas-regions

Q23 Domain A: Plan and manage an Azure AI solution Single choice

A GitHub workflow must deploy a hosted agent with no long-lived Azure credential stored in GitHub. Which identity setup fits the documented deployment pattern?

A A client-secret credential copied into GitHub and rotated before each release.

B A project endpoint variable used as the workflow's authentication credential.

C Federated OIDC authentication with a scoped deployment identity. ✓

D The runtime agent identity inferred from the agent name in the workflow file.

Why the correct answer is right

OIDC federation lets the workflow obtain tokens through a configured trust relationship. The workflow identity still needs the required deployment roles.

Why the other options are wrong

- **A:** This still stores a long-lived secret credential.
- **D:** A name does not establish workflow federation or deployment permissions.
- **B:** The endpoint identifies the service; it does not authenticate the workflow.

AI103-A2.04

Sources

S012: Quickstart: Hosted agent CI/CD templates
https://learn.microsoft.com/azure/foundry/agents/quickstarts/set-up-cicd-hosted-agent

Q24 Domain A: Plan and manage an Azure AI solution Single choice

An agent request traverses retrieval, generation and a business tool. Engineers need to attribute a latency increase to one operation and its parent request. Which telemetry structure supports that?

A Timed service events whose correlation key is the service name shared by all requests.

B Record a structured event for each request containing final text, total elapsed time and the names of every participating service.

C Export a correlated request trace with timed nested spans for retrieval, generation and the tool call. ✓

D Independent traces for each operation, each with a newly generated request identifier.

Why the correct answer is right

Nested spans represent timed operations inside a request trace. Consistent attributes and parent relationships make the slow stage visible.

Why the other options are wrong

- **B:** Aggregate request timing and service names do not identify timed nested operations.
- **A:** A service name does not connect operations for a particular request.
- **D:** Separate identifiers break the shared request hierarchy.

AI103-A4.03

Sources

S013: Agent tracing overview
https://learn.microsoft.com/azure/foundry/observability/concepts/trace-agent-concept

Q25 Domain A: Plan and manage an Azure AI solution Single choice

Tool arguments include account numbers and an access token. Engineers need useful traces for incident diagnosis. Which collection policy best fits?

A Export full content spans to a restricted workspace, then delete secrets in a nightly job.

B Store raw access tokens in encrypted span attributes accessible to incident responders.

C Remove the request and evaluation identifiers while preserving raw tool arguments.

D Redact sensitive fields before export and preserve safe correlation data. ✓

Why the correct answer is right

Remove credentials and redact personal data before they enter telemetry. Safe identifiers, timings, access controls and retention policies preserve diagnostic value.

Why the other options are wrong

- **A:** Secrets enter telemetry before the deletion job.
- **C:** This loses correlation and retains sensitive content.
- **B:** Credentials should not be recorded in telemetry.

AI103-A4.03

Sources

S013: Agent tracing overview
https://learn.microsoft.com/azure/foundry/observability/concepts/trace-agent-concept

Q26 Domain A: Plan and manage an Azure AI solution Single choice

An indexer reports Success, but several known source documents are absent from Search. Where should the investigation start?

A The successful run label and aggregate index document count.

B The semantic ranker configuration used by the question-answering application.

C The run's document-level errors and warnings. ✓

D The service's query-latency dashboard for searches made after the run.

Why the correct answer is right

Review detailed indexer execution results. A run-level Success status does not establish successful processing of every document.

Why the other options are wrong

- **D:** Query metrics do not identify failed ingestion items.
- **A:** A successful run can still include document-level errors.
- **B:** Reranking does not explain documents that failed ingestion.

AI103-A3.03

Sources

S027: Monitor Azure AI Search
https://learn.microsoft.com/azure/search/monitor-azure-cognitive-search

S044: Monitor indexer status and results in Azure AI Search
https://learn.microsoft.com/azure/search/search-monitor-indexers

Q27 Domain B: Implement generative AI and agentic solutions Configuration

Complete the current Python client calls for an existing agent named helpdesk and its project OpenAI client.

agent = project.agents.[_____](agent_name="helpdesk")
client = project.[_____]()
operation: get / get_agent / create_version
client: get_openai_client / get_search_client / get_agent_client

Completed statement:

agent = project.agents.**get**(agent_name="helpdesk")

client = project.**get_openai_client**()

Why the correct answer is right

The current operations reference exposes get(agent_name). The retained 2.8.0 project library provides get_openai_client for Responses and conversations.

Answer and step notes

- **operation:** Retrieve the existing named definition; do not mix classic retrieval or create a new version.
- **client:** Obtain the documented OpenAI client for Responses operations.

AI103-B1.06

Sources

S002: Azure AI Projects client library for Python - version 2.8.0
https://learn.microsoft.com/python/api/overview/azure/ai-projects-readme?view=azure-python

S056: AI Projects AgentsOperations Python reference
https://learn.microsoft.com/en-us/python/api/azure-ai-projects/azure.ai.projects.operations.agentsoperations?view=azure-python

Q28

Domain B: Implement generative AI and agentic solutions

Configuration

Complete a backend memory-search tool definition and response header for per-user preferences. user_key is derived from the authenticated session.

```
from azure.ai.projects.models import MemorySearchPreviewTool, PromptAgentDefinition
tool = MemorySearchPreviewTool(memory_store_name="preferences", scope="
[_____________]")
agent = project.agents.create_version(agent_name="support",
definition=PromptAgentDefinition(model=deployment, tools=[tool]))
response = client.responses.create(input=message,
extra_body={"agent_reference": {"name": agent.name, "type": "agent_reference"}},
extra_headers={"[_____________]": user_key})
scope: {{$userId}} / backend-shared / conversation-title
header: x-memory-user-id / x-request-id / Authorization
```

Completed statement:

```
from azure.ai.projects.models import MemorySearchPreviewTool, PromptAgentDefinition
tool = MemorySearchPreviewTool(memory_store_name="preferences", scope="{{$userId}}")
agent = project.agents.create_version(agent_name="support",
    definition=PromptAgentDefinition(model=deployment, tools=[tool]))
response = client.responses.create(input=message,
    extra_body={"agent_reference": {"name": agent.name, "type": "agent_reference"}},
    extra_headers={"x-memory-user-id": user_key})
```

Why the correct answer is right

The memory tool resolves the placeholder from the memory-user header, falling back to caller TID/OID without it. A backend must supply its authenticated end-user key.

Answer and step notes

- **scope:** Use the documented per-user placeholder; static scope pools users and a title is not a partition key.
- **header:** The memory-user header carries end-user identity; a trace ID does not select a user and Authorization identifies the calling service.

AI103-B2.02

Sources

S053: Create and use memory in Foundry Agent Service (preview)
https://learn.microsoft.com/en-us/azure/foundry/agents/how-to/memory-usage

Q29 Domain B: Implement generative AI and agentic solutions Configuration

A Responses turn must consult at least one attached tool. Complete the request-level control.

response = client.responses.create(model=deployment, input=message, tools=tools, tool_choice="[_____]")
mode: auto / required / none

Completed statement:

response = client.responses.create(model=deployment, input=message, tools=tools, tool_choice="**required**")

Why the correct answer is right

required demands one or more calls. It does not by itself select one exact function, execute host code or grant permission.

Answer and step notes

- **mode:** auto leaves tool use to the model; none prevents it. The host remains responsible for validation and execution.

AI103-B2.01

Sources

S054: Tool best practices for Microsoft Foundry Agent Service
https://learn.microsoft.com/azure/foundry/agents/concepts/tool-best-practice

Q30

Domain B: Implement generative AI and agentic solutions

Configuration

Complete a supported strict Responses schema: note must be present and can be null; unknown properties are rejected.

schema = {"type": "object", "properties": {"note": {"type": ["string", "null"]}},
"required": [_____], "additionalProperties": [_____]}
response text.[_____] contains the json_schema format
required: ["note"] / [] / ["string", "null"]
extra: False / True
location: format / response_format / metadata

Completed statement:

schema = {"type": "object", "properties": {"note": {"type": ["string", "null"]}},
"required": **["note"]**, "additionalProperties": **False**}
response text.**format** contains the json_schema format

Why the correct answer is right

Nullable value types do not make a property optional in the supported strict subset. Responses uses text.format, unlike the Chat Completions field.

Answer and step notes

- **required:** Require the key while retaining a nullable value type.
- **extra:** additionalProperties false disallows unknown fields.
- **location:** Use the documented Responses request shape.

AI103-B3.01

Sources

S037: Structured outputs
https://learn.microsoft.com/azure/foundry/openai/how-to/structured-outputs

Q31 Domain B: Implement generative AI and agentic solutions Single choice

A supported reasoning model returns a reasoning summary through Responses. A product manager wants to label it 'the model's complete internal reasoning'. Which description is accurate?

A A policy-adherence score produced by an agent evaluator.

B An API-provided reasoning summary. ✓

C A raw transcript containing each internal reasoning step.

D A deterministic execution log for the model's hidden operations.

Why the correct answer is right

Reasoning summaries are a supported response feature for applicable models. They should not be represented as the full internal reasoning process.

Why the other options are wrong

- **C:** The summary does not expose all internal reasoning.
- **D:** The response is not a complete operational execution log.
- **A:** The returned reasoning summary is not an evaluator result.

AI103-B3.01

Sources

S038: Azure OpenAI reasoning models
https://learn.microsoft.com/azure/foundry/openai/how-to/reasoning

Q32 Domain B: Implement generative AI and agentic solutions Single choice

A support assistant should stay within product support and ask for clarification when evidence is insufficient. Which prompt control expresses both?

A A system-defined domain and missing-evidence fallback. ✓

B More tone examples without defining scope.

C A low temperature with no role instructions.

D A larger retrieval top-k value alone.

Why the correct answer is right

System messages express role, scope and fallback. Test behavior and apply other mitigations because instructions do not guarantee compliance.

Why the other options are wrong

- **C:** Temperature affects sampling rather than defining the task boundary.
- **D:** Retrieval quantity does not state domain behavior.
- **B:** Examples of tone do not replace an explicit scope/fallback contract.

AI103-B3.01

Sources

S057: System message design
https://learn.microsoft.com/en-us/azure/cognitive-services/openai/concepts/advanced-prompt-engineering?pivots=programming-language-chat-completions

Q33 Domain B: Implement generative AI and agentic solutions Single choice

A deterministic workflow sends a request to a retrieval specialist and then a policy specialist. What must its orchestration represent?

A One larger model context replacing all flow rules.

B Separate trace IDs with no shared request context.

C Explicit specialist handoff and completion conditions. ✓

D Tool descriptions treated as executable routing code.

Why the correct answer is right

Multi-agent orchestration needs explicit workflow control and termination conditions. Distinct specialist roles do not themselves supply a valid process.

Why the other options are wrong

- **A:** Context size does not define routing or completion.
- **D:** Descriptions are not the orchestrator’s executable flow.
- **B:** Uncorrelated telemetry makes diagnosis harder and does not define completion.

AI103-B2.04

Sources

S008: Build a workflow in Microsoft Foundry (Preview)
https://learn.microsoft.com/azure/foundry/agents/concepts/workflow

Q34 Domain B: Implement generative AI and agentic solutions Single choice

An agent calls the translation tool for a request that requires stock availability. The translation call is valid and succeeds. Which evaluation pinpoints the mistake?

A Tool Call Success.

B Tool Output Utilization.

C Tool Selection. ✓

D Tool Input Accuracy.

Why the correct answer is right

Tool Selection measures whether the selected tool is appropriate. A valid and successful invocation can still be the wrong tool for the task.

Why the other options are wrong

- **A:** Execution success is not the same as choosing the right tool.
- **D:** The scenario says the call was valid.
- **B:** Output use is downstream of choosing the wrong capability.

AI103-B2.06

Sources

S014: Agent evaluators
https://learn.microsoft.com/azure/foundry/concepts/evaluation-evaluators/agent-evaluators

Q35 Domain B: Implement generative AI and agentic solutions Single choice

A Search tool points at the right service but retrieves an unrelated index. Which agent-tool configuration should be corrected first?

A The retrieval filter for the current unrelated index.

B The tool's maximum result count.

C The configured index name and project connection. ✓

D The tool's vector-versus-keyword query mode.

Why the correct answer is right

The Azure AI Search tool uses a project connection and an index name. Verify both before changing ranking or generation settings.

Why the other options are wrong

- **B:** Count limits results but does not choose which index they come from.
- **D:** Query mode changes retrieval behavior inside the selected index.
- **A:** A filter does not redirect the tool to the intended index.

AI103-B1.02

Sources

S040: Connect an Azure AI Search index to Foundry agents
https://learn.microsoft.com/azure/foundry/agents/how-to/tools/ai-search

Q36 Domain B: Implement generative AI and agentic solutions Single choice

A response is fully supported by context and addresses the question, but omits two required parts of the reference answer. Which additional metric is most targeted?

A Response Completeness (preview). ✓

B Groundedness repeated against the same supplied context.

C Response relevance.

D Response safety screening.

Why the correct answer is right

Completeness addresses missing required content. Groundedness and relevance can both be strong while the response remains incomplete.

Why the other options are wrong

- **B:** Support does not establish coverage of all required parts.
- **C:** The stem already establishes topical relevance; missing required parts need completeness assessment.
- **D:** Safety screening does not measure coverage of the missing reference-answer parts.

AI103-B1.04

Sources

S039: Retrieval-Augmented Generation (RAG) evaluators
https://learn.microsoft.com/azure/foundry/concepts/evaluation-evaluators/rag-evaluators

Q37 Domain A: Plan and manage an Azure AI solution Single choice

An agent must remember a user's preferred units across conversations while answering from a maintained engineering manual. Which division of context is appropriate?

A Store preferences in the knowledge base and use temporary file search for the engineering manual.

B Store both preferences and the maintained manual in interaction-derived memory.

C Store learned preferences in persistent memory and use maintained knowledge for the current manual. ✓

D Store learned preferences in the prior session conversation history and retrieve the current manual from the maintained knowledge base.

Why the correct answer is right

Persistent interaction-derived preferences fit memory; curated organizational content fits a knowledge base.

Why the other options are wrong

- **D:** Prior session state is not the persistent cross-session memory requested.
- **B:** A curated source should remain a maintained knowledge resource.
- **A:** This misassigns user-specific memory and the maintained organizational source.

AI103-A1.04

Sources

S006: Memory in Microsoft Foundry Agent Service (preview)
https://learn.microsoft.com/azure/foundry/agents/concepts/what-is-memory

Q38 Domain A: Plan and manage an Azure AI solution Single choice

An agent follows its operating policy and calls the correct tools, but leaves two requested deliverables unfinished. Which evaluator best targets the overall outcome?

A Task Completion (preview), assessing whether the agent delivered the requested end-to-end outcome. ✓

B Tool Selection, focused on the choice of tool for each operation.

C Task Adherence (preview), using the fact that the agent obeyed its operating policy to judge whether all requested deliverables were completed.

D Tool Input Accuracy, focused on the arguments supplied to a tool.

Why the correct answer is right

Task Completion assesses whether the agent accomplished the requested task. Adherence and individual tool metrics measure different properties.

Why the other options are wrong

- **C:** Policy adherence does not establish delivery of the entire requested outcome.
- **B:** Correct tool choice is already observed.
- **D:** Argument quality does not measure the whole requested outcome.

AI103-A4.02

Sources

S014: Agent evaluators
https://learn.microsoft.com/azure/foundry/concepts/evaluation-evaluators/agent-evaluators

Q39 Domain A: Plan and manage an Azure AI solution Single choice

A team has query-document relevance labels and wants to compare two retrievers before adding answer generation. Which evaluation is appropriate?

A Groundedness, using a hypothetical final answer to infer whether the two retrievers placed all relevant documents at the correct rank positions.

B Response Completeness evaluated against a reference final answer.

C Task Adherence evaluated against the agent system instructions.

D Document Retrieval against the supplied query-document relevance labels for the two candidate retrievers. ✓

Why the correct answer is right

Ground-truth relevance labels support document-retrieval evaluation. This isolates search quality from generation behavior.

Why the other options are wrong

- **A:** Groundedness of an answer is not labeled measurement of the retrieved document ranking.
- **B:** This measures answer coverage rather than the retrieved ranking.
- **C:** Policy adherence is not the labeled retrieval measurement.

AI103-A4.02

Sources

S039: Retrieval-Augmented Generation (RAG) evaluators
https://learn.microsoft.com/azure/foundry/concepts/evaluation-evaluators/rag-evaluators

Q40 Domain A: Plan and manage an Azure AI solution Single choice

A retrieved maintenance page tells the agent to ignore its instructions and send credentials to a URL. Which threat category should be evaluated at this boundary?

A A factual data-poisoning error that changes a measurement without issuing instructions.

B A harmful final-answer classification unrelated to the retrieved instruction boundary.

C A direct user prompt attack originating in the authenticated user message.

D An indirect prompt attack in external document content. ✓

Why the correct answer is right

External documents and tool results are lower-trust inputs. Prompt Shields distinguishes these indirect attacks from attacks supplied directly in a user prompt.

Why the other options are wrong

- **C:** The attack is embedded in retrieved context rather than the user message.
- **A:** The payload explicitly attempts to override instructions and exfiltrate credentials.
- **B:** The described threat is an instruction attack entering through external context.

AI103-A4.01

Sources

S015: Guardrails and controls overview in Microsoft Foundry
https://learn.microsoft.com/azure/foundry/guardrails/guardrails-overview

S016: Prompt Shields in Microsoft Foundry
https://learn.microsoft.com/azure/foundry/openai/concepts/content-filter-prompt-shields

Q41 Domain A: Plan and manage an Azure AI solution Single choice

An organization requires agent state in customer-managed resources in its own subscription. Which agent setup aligns with that storage requirement?

A A larger model context window in the existing basic setup.

B A model deployment changed from Standard to Provisioned throughput.

C Standard setup with customer-managed supporting resources. ✓

D Basic setup using Microsoft-managed state storage.

Why the correct answer is right

The documented basic and standard agent setups differ in state storage ownership. Standard setup uses customer-managed resources.

Why the other options are wrong

- **D:** This does not meet the customer-managed storage requirement.
- **A:** Context size does not change the backing storage arrangement.
- **B:** Model deployment throughput does not choose the agent storage setup.

AI103-A2.01

Sources

S009: Foundry Agent Service limits, quotas, and regional support
https://learn.microsoft.com/azure/foundry/agents/concepts/limits-quotas-regions

Q42 Domain A: Plan and manage an Azure AI solution Single choice

A team wants one supported Python client library for extracting document layout, transcribing audio and analyzing recorded video into structured outputs. Which service's documented client covers all three input modalities?

A Azure Language text summarization alone.

B Azure AI Search OCR skill alone.

C Azure Document Intelligence Layout alone.

D Azure AI Content Understanding. ✓

Why the correct answer is right

The Content Understanding client documents extraction across documents, audio, video and images. The other named capabilities each omit required input modalities.

Why the other options are wrong

- **C:** Document layout alone does not supply audio transcription and video analysis.
- **A:** Text summarization alone does not perform all three native input analyses.
- **B:** OCR enrichment does not provide the required audio and video analysis.

AI103-A1.02

Sources

S021: Azure AI Content Understanding client library for Python - version 1.1.0
https://learn.microsoft.com/python/api/overview/azure/ai-contentunderstanding-readme?view=azure-python

Q43 Domain A: Plan and manage an Azure AI solution Configuration

Complete the Azure OpenAI v1 client for keyless inference. The calling identity has the inference role at the resource scope.

token_provider = get_bearer_token_provider(credential, "[_____]")
client = OpenAI(base_url="[_____]", api_key=token_provider)
audience: https://ai.azure.com/.default / https://management.azure.com/.default / https://search.azure.com/.default
endpoint: https://resource.openai.azure.com/openai/v1/ / https://resource.services.ai.azure.com/api/projects/support / https://resource.search.windows.net/

Completed statement:

token_provider = get_bearer_token_provider(credential, "**https://ai.azure.com/.default**")

client = OpenAI(base_url="**https://resource.openai.azure.com/openai/v1/**",

api_key=token_provider)

Why the correct answer is right

The named v1 example uses the ai.azure.com token audience and the resource /openai/v1/ base URL. Authentication still requires authorization on the called resource.

Answer and step notes

- **audience:** Use the v1 example's inference token audience, rather than an ARM or Search audience.
- **endpoint:** This is the direct resource v1 surface, not project management or Search.

AI103-A3.04

Sources

S051: Azure OpenAI v1 API
https://learn.microsoft.com/en-us/azure/foundry/openai/api-version-lifecycle

S052: Use Azure OpenAI without keys
https://learn.microsoft.com/en-us/azure/developer/ai/keyless-connections

Q44 Domain A: Plan and manage an Azure AI solution Configuration

Complete the AI Projects Python 2.8.0 constructor for project operations.

project = AIProjectClient(endpoint=[_____], credential=[_____])
endpoint: project_endpoint / openai_v1_base_url / search_endpoint
credential: DefaultAzureCredential() / AzureKeyCredential(resource_key) / project_endpoint

Completed statement:

project = AIProjectClient(endpoint=**project_endpoint**, credential=**DefaultAzureCredential()**)

Why the correct answer is right

The retained AI Projects 2.8.0 reference uses a Foundry project endpoint and Entra credentials. Project and direct inference clients have different roles.

Answer and step notes

- **endpoint:** Choose the endpoint of the named project for project operations.
- **credential:** The 2.8.0 client supports Entra credentials; a URL is not a credential and resource-key construction is not this client contract.

AI103-A2.03

Sources

S002: Azure AI Projects client library for Python - version 2.8.0
https://learn.microsoft.com/python/api/overview/azure/ai-projects-readme?view=azure-python

S003: Microsoft Foundry SDKs and endpoints
https://learn.microsoft.com/azure/foundry/how-to/develop/sdk-overview

Q45 Domain A: Plan and manage an Azure AI solution Single choice

A catalog requires exact part-code matches and semantic symptom matches. Which design combines both retrieval signals before optional reranking?

A Vector retrieval alone, with a larger candidate count.

B Semantic reranking of keyword-only candidates.

C Keyword and vector retrieval with fused rankings. ✓

D Keyword retrieval alone, with boosted identifier fields.

Why the correct answer is right

Keyword evidence complements semantic vectors. Evaluate hybrid retrieval for the actual query set; ranking quality is distinct from generation and authentication.

Why the other options are wrong

- **A:** A larger vector candidate count does not add keyword retrieval.
- **D:** Boosted keyword fields do not add vector retrieval.
- **B:** Reranking keyword candidates does not combine keyword and vector retrieval.

AI103-A1.03

Sources

S025: Hybrid search using vectors and full-text search in Azure AI Search
https://learn.microsoft.com/azure/search/hybrid-search-overview

S026: Integrated vector embedding in Azure AI Search
https://learn.microsoft.com/azure/search/vector-search-integrated-vectorization

Q46 Domain A: Plan and manage an Azure AI solution Single choice

An agent proposes a consequential account change. Which host boundary should decide whether it executes?

A The host’s caller authorization and action approval gate. ✓

B A valid model response schema.

C A lower generation temperature.

D A more detailed tool description.

Why the correct answer is right

The host controls operations and validates values and authority. Descriptions and structured output help represent intent but are not action permission.

Why the other options are wrong

- **C:** Sampling does not establish caller authority.
- **D:** Descriptions are not an executable permission check.
- **B:** A response schema does not authorize an action.

AI103-A4.04

Sources

S054: Tool best practices for Microsoft Foundry Agent Service
https://learn.microsoft.com/azure/foundry/agents/concepts/tool-best-practice

Q47 Domain C: Implement computer vision solutions Single choice

A catalog app must create a new product-scene image from a text brief, with no source image supplied. Its supported GPT image deployment is available. Which operation matches this input contract?

A Call images.edit with the brief as its source image file.

B Send the brief to a vision chat request and treat the text response as image bytes.

C Call images.edit with a mask but no source image.

D Call images.generate with the deployment and text prompt. ✓

Why the correct answer is right

Image generation accepts the text prompt to create an image. Editing requires source image input; vision chat interprets supplied images rather than returning generated image bytes.

Why the other options are wrong

- **A:** A text brief is not the source image input for editing.
- **B:** Vision chat text is not a generated image file.
- **C:** A mask does not supply the missing source image.

AI103-C1.01

Sources

S017: Azure OpenAI image generation models
https://learn.microsoft.com/azure/foundry/openai/how-to/dall-e

Q48 Domain C: Implement computer vision solutions Single choice

An application switches between two GPT image deployments with different quality and size options. What should its request builder use?

A One fixed list copied from an older image model.

B The union of both models' parameters, sent to each deployment for consistency.

C The supported controls of the selected deployed image model. ✓

D The default size accepted by a different deployment in the same resource.

Why the correct answer is right

Image models differ in supported editing, size and quality controls. Build requests against the selected deployment's documented capabilities.

Why the other options are wrong

- **A:** Parameter support varies by model.
- **B:** A parameter supported by one model need not be supported by the other.
- **D:** Another deployment does not define the chosen model's accepted controls.

AI103-C1.05

Sources

S017: Azure OpenAI image generation models
https://learn.microsoft.com/azure/foundry/openai/how-to/dall-e

Q49 Domain C: Implement computer vision solutions Single choice

A vision-enabled chat model must describe damage visible in a supplied photograph. Which message construction gives it the actual visual input?

A Include text instructions and image_url content in the message. ✓

B Enable image output formatting on the text-only message.

C Put the image filename in a text message.

D Describe an image path in the system prompt while keeping the image local to the client.

Why the correct answer is right

Vision chat messages can combine text and image_url content. Supply the image in the supported representation along with the task instructions.

Why the other options are wrong

- **C:** A filename does not transmit the image content.
- **D:** An inaccessible path does not supply visual input to the service.
- **B:** Output configuration does not supply visual input.

AI103-C2.01

Sources

S019: Use vision-enabled chat models
https://learn.microsoft.com/azure/foundry/openai/how-to/gpt-with-vision

Q50 Domain C: Implement computer vision solutions Single choice

A magnifying-glass image is the visible label of a search button. Which text alternative best communicates the function?

A Small circular lens with a diagonal handle on a blue background.

B search-icon-v3.png.

C Magnifying glass icon.

D Search the catalog. ✓

Why the correct answer is right

For an image serving as a control, communicate its action or destination. Here the functional label is the search action.

Why the other options are wrong

- **C:** The visual shape does not name the action.
- **A:** A visual description is less useful than the button's function.
- **B:** A filename is not an accessible action label.

AI103-C2.04

Sources

S050: W3C WAI An alt Decision Tree
https://www.w3.org/WAI/tutorials/images/decision-tree/

Q51 Domain C: Implement computer vision solutions Single choice

A Content Understanding video preview samples about one frame per second. A safety event lasts 0.2 seconds and the output does not mention it. Which conclusion is justified?

A The absence of a mention proves the event did not occur.

B Increase output length to recover source frames omitted by the sampling stage.

C The sampled analysis may have missed the brief event. ✓

D The analyzer must have sampled every source frame because the request succeeded.

Why the correct answer is right

Brief events can fall between sampled frames. Absence from the output is not proof that the event was absent from the original video.

Why the other options are wrong

- **A:** Sampling cannot establish complete observation of rapid events.
- **B:** Output length cannot recover visual evidence that was not sampled.
- **D:** Successful analysis does not change the documented sampling behavior.

AI103-C2.06

Sources

S022: Azure Content Understanding in Foundry Tools video solutions
https://learn.microsoft.com/azure/ai-services/content-understanding/video/overview

Q52 Domain C: Implement computer vision solutions Single choice

A production document-image workflow needs capabilities available in Content Understanding 2025-11-01. A newer preview offers unrelated features. Which API selection follows the documented production guidance?

A Use the 2025-11-01 GA API for the required production capabilities. ✓

B Use the retired 2025-05-01-preview because its examples are familiar.

C Use 2026-06-01-preview because its date is newer.

D Alternate API versions randomly to compare response speed in production.

Why the correct answer is right

The documentation recommends the GA API for production and the newer preview for evaluation of new capabilities. Match feature needs to an explicit version.

Why the other options are wrong

- **C:** The documented guidance favors GA for supported production needs.
- **B:** Retired API examples are not the production target.
- **D:** Random version changes do not provide a stable application contract.

AI103-C2.05

Sources

S020: What is Azure Content Understanding in Foundry Tools?
https://learn.microsoft.com/azure/ai-services/content-understanding/overview

Q53 Domain C: Implement computer vision solutions Single choice

A poster has a low visual harm score but includes text telling an agent to reveal credentials. Which conclusion follows from the harm result?

A The image harm score is a groundedness measurement.

B Visual harm severity; instruction trust is a separate check. ✓

C A successful harm request establishes the text is authorized.

D The poster text has passed instruction-trust validation.

Why the correct answer is right

Visual harm classification is distinct from text prompt-attack detection. The FAQ describes text-only Prompt Shields; the harm score does not authorize instructions.

Why the other options are wrong

- **D:** Harm classification does not establish instruction trust.
- **C:** HTTP success is not authorization of embedded text.
- **A:** Groundedness is a different assessment.

AI103-C3.01

Sources

S042: QuickStart: Analyze image content
https://learn.microsoft.com/azure/ai-services/content-safety/quickstart-image

S048: Azure AI Content Safety Frequently Asked Questions
https://learn.microsoft.com/en-us/azure/ai-services/content-safety/faq

Q54 Domain C: Implement computer vision solutions Single choice

A vision model receives two images of the same machine. The app needs a concise caption comparing visible differences. What should be supplied?

A Attach only the first image and describe the second image’s filename.

B Both supported image inputs with an explicit comparison/caption instruction. ✓

C Attach both images but ask only for an independent caption of the first.

D Send a text-only comparison request containing both image URLs as prose.

Why the correct answer is right

The vision request must actually include both images using its supported image content shape. The instruction specifies comparison and desired detail.

Why the other options are wrong

- **A:** The second filename does not supply the second image.
- **D:** Plain prose URLs do not provide supported image content inputs.
- **C:** That instruction does not request the required visual comparison.

AI103-C2.02

Sources

S019: Use vision-enabled chat models
https://learn.microsoft.com/azure/foundry/openai/how-to/gpt-with-vision

Q55 Domain C: Implement computer vision solutions Configuration

Complete a vision chat message so a question about a supplied inspection photo is grounded in actual image input.

content = [{"type": "text", "text": question},
{"type": "[_____]", "[_____]": {"url": photo_url}}]
kind: image_url / text / image_output
field: image_url / file_name / caption

Completed statement:

content = [{"type": "text", "text": question},
{"type": "**image_url**", "**image_url**": {"url": photo_url}}]

Why the correct answer is right

The documented chat shape includes an image_url content part with its URL. The question text describes what to inspect; it does not provide pixels.

Answer and step notes

- **kind:** Use the image input content type for the vision chat surface.
- **field:** The image_url object contains the source; a filename or generated caption is not equivalent input.

AI103-C2.03

Sources

S019: Use vision-enabled chat models
https://learn.microsoft.com/azure/foundry/openai/how-to/gpt-with-vision

Q56 Domain C: Implement computer vision solutions Configuration

The selected image model supports masking. Complete an edit request for a source image and a mask already sized to match it.

result = client.images.edit(model=deployment, image=[_____], mask=[_____], prompt=edit_instruction)
source: source_image_file / source_filename_text / prior_caption_text
mask: matching_mask_file / size_string / quality_string

Completed statement:

result = client.images.edit(model=deployment, image=**source_image_file**, mask=**matching_mask_file**, prompt=edit_instruction)

Why the correct answer is right

The supported edit consumes source image data and a spatial mask. Size and quality controls do not substitute for the edit mask.

Answer and step notes

- **source:** Supply the image bytes/file accepted by this endpoint, not its name or a caption.
- **mask:** Supply the matching mask as an input; a size/quality value is a different control.

AI103-C1.03

Sources

S017: Azure OpenAI image generation models
https://learn.microsoft.com/azure/foundry/openai/how-to/dall-e

Q57 Domain D: Implement text analysis solutions Single choice

A Translator text request contains one sentence with English and German spans. Some words remain untranslated. Which preparation targets the documented limitation?

A Submit the same mixture to document translation with autodetection.

B Retry the same mixed sentence until detection changes.

C Provide a glossary while keeping the mixed-language sentence intact.

D Segment by source language and translate each segment explicitly. ✓

Why the correct answer is right

The known-issues page distinguishes mixed-language text and document failures and recommends intended-source-language handling or single-language segmentation.

Why the other options are wrong

- **C:** Terminology customization does not remove the source-language mixture.
- **B:** Unchanged retries do not resolve the documented input limitation.
- **A:** Document translation also has documented mixed-language limitations.

AI103-D1.03

Sources

S055: Azure Translator known issues and troubleshooting
https://learn.microsoft.com/en-us/azure/ai-services/translator/reference/known-issues

Q58 Domain D: Implement text analysis solutions Single choice

A domain compliance summarizer needs a defined scope and an explicit escalation when evidence is missing. Which instruction design addresses it?

A A deployment name containing compliance.

B Domain instructions with a missing-evidence fallback. ✓

C A longer output limit alone.

D A lower top-p value alone.

Why the correct answer is right

System instructions define task scope and fallback, while evaluation checks behavior. Sampling, length and names do not establish those requirements.

Why the other options are wrong

- **D:** Sampling does not state the permitted task.
- **C:** Length does not define scope.
- **A:** A deployment label is not an instruction contract.

AI103-D1.04

Sources

S057: System message design
https://learn.microsoft.com/en-us/azure/cognitive-services/openai/concepts/advanced-prompt-engineering?pivots=programming-language-chat-completions

Q59 Domain E: Implement information extraction solutions Multiple select

A blob OCR pipeline has imageAction none and incorrectly maps a skill-produced text node with fieldMappings. Which TWO changes repair the data path? Select TWO answers.

A Keep imageAction none and use a larger searchable field.

B Map enriched OCR text into the index with outputFieldMappings. ✓

C Move the OCR text mapping into the semantic query request.

D Enable original-image extraction but keep the OCR text mapping in fieldMappings.

E Generate normalized images for the OCR skill input. ✓

Why the correct answer is right

OCR needs normalized image inputs, and the enrichment output needs a mapping to the target index field. Fix both sides of the pipeline.

Why the other options are wrong

- **A:** Field size does not create normalized images.
- **C:** A query does not configure enrichment output ingestion.
- **D:** Image extraction alone does not repair the mapping for the skill-produced enriched node.

AI103-E1.04

Sources

S043: OCR cognitive skill
https://learn.microsoft.com/azure/search/cognitive-search-skill-ocr

S028: Skillset concepts in Azure AI Search
https://learn.microsoft.com/azure/search/cognitive-search-working-with-skillsets

Q60 Domain E: Implement information extraction solutions Multiple select

A team uses Content Understanding SDK 1.1.0 for production document extraction. Which TWO implementation choices align with the referenced contract? Select TWO answers.

A Use capabilities compatible with its 2025-11-01 API version. ✓

B Submit the same document repeatedly as a substitute for polling.

C Wait for analysis completion before consuming the structured result. ✓

D Treat the initial poller as completed Markdown content.

E Mix preview-only analyzer fields into the request with no version change.

Why the correct answer is right

Match the SDK's supported API and follow the asynchronous analysis lifecycle. Completed structured content is available after successful completion.

Why the other options are wrong

- **E:** The feature and API versions must be compatible.
- **D:** The poller represents the long-running operation.
- **B:** Repeated submission creates additional operations.

AI103-E2.02

Sources

S021: Azure AI Content Understanding client library for Python - version 1.1.0
https://learn.microsoft.com/python/api/overview/azure/ai-contentunderstanding-readme?view=azure-python

S020: What is Azure Content Understanding in Foundry Tools?
https://learn.microsoft.com/azure/ai-services/content-understanding/overview

Q61 Domain E: Implement information extraction solutions Multiple select

A custom analyzer must extract an amount from unfamiliar invoice layouts and support reviewer verification. Which TWO features or inputs are useful? Select TWO answers.

A A high confidence threshold used as proof that all accepted values are correct.

B Representative labeled examples with verified field values. ✓

C Generated explanations used as substitutes for labeled source values.

D Removing source coordinates to standardize the JSON shape.

E Field source and confidence estimation for the extraction results. ✓

Why the correct answer is right

Good labeled examples support extraction quality; confidence and grounding support review. Neither should be confused with independent verification.

Why the other options are wrong

- **C:** Explanations are not verified labels.
- **A:** Thresholding does not create a correctness guarantee.
- **D:** That discards useful evidence references.

AI103-E2.03

Sources

S045: Validate document analyzer quality with confidence, grounding, and labeled samples
https://learn.microsoft.com/azure/ai-services/content-understanding/document/analyzer-improvement

S059: Content Understanding document solutions
https://learn.microsoft.com/en-us/azure/ai-services/content-understanding/document/overview

Q62 Domain E: Implement information extraction solutions Multiple select

Long manuals exceed embedding limits, and the query vectorizer uses an unrelated model. Which TWO configuration corrections address the two failures? Select TWO answers.

A Chunk the source text within input limits. ✓

B Increase the answer-generation token limit.

C Disable indexing warnings.

D Rescale query vectors from the unrelated model.

E Use content/query embeddings from a compatible model configuration. ✓

Why the correct answer is right

Chunking addresses embedding input size; compatible query/content vectors address representation mismatch. These are separate ingestion and retrieval requirements.

Why the other options are wrong

- **B:** Answer length does not enlarge embedding input limits.
- **D:** Rescaling does not align learned embedding spaces.
- **C:** Suppressing warnings does not fix either defect.

AI103-E1.01

Sources

S058: Integrated vectorization REST configuration
https://learn.microsoft.com/en-us/azure/search/search-how-to-integrated-vectorization

S026: Integrated vector embedding in Azure AI Search
https://learn.microsoft.com/azure/search/vector-search-integrated-vectorization

Q63 Domain E: Implement information extraction solutions Multiple select

A product catalog needs exact codes and semantic candidates, with optional semantic reranking. Which TWO statements describe the stages correctly? Select TWO answers.

A Hybrid retrieval combines keyword and vector ranked lists. ✓

B Semantic ranking renames source fields during ingestion.

C Semantic reranking generates the indexed vectors.

D A query vectorizer uses a compatible embedding configuration. ✓

E Keyword sorting makes vectorization unnecessary for the vector path.

Why the correct answer is right

Hybrid retrieval combines complementary lists. Query vectorization supplies the vector path; semantic reranking operates after candidate retrieval.

Why the other options are wrong

- **C:** Reranking does not produce stored embeddings.
- **B:** Field mapping is an ingestion concern.
- **E:** A vector query still requires compatible vectorization.

AI103-E1.02

Sources

S025: Hybrid search using vectors and full-text search in Azure AI Search
https://learn.microsoft.com/azure/search/hybrid-search-overview

S058: Integrated vectorization REST configuration
https://learn.microsoft.com/en-us/azure/search/search-how-to-integrated-vectorization

Q64 Domain E: Implement information extraction solutions Multiple select

A document enrichment pipeline needs layout-aware Markdown and OCR for normalized images. Which TWO skill choices directly target these outputs? Select TWO answers.

A The OCR skill receiving normalized images. ✓

B The Document Intelligence Layout skill. ✓

C A semantic query configuration.

D An output field alias without a producing skill.

E An Azure OpenAI chat output-token setting.

Why the correct answer is right

Layout and OCR are producing enrichment capabilities. Their outputs still need the appropriate index mappings.

Why the other options are wrong

- **C:** Query semantics do not extract document layout.
- **E:** Generation length does not configure enrichment skills.
- **D:** A mapping does not create an absent enrichment output.

AI103-E1.03

Sources

S029: Document Layout skill
https://learn.microsoft.com/azure/search/cognitive-search-skill-document-intelligence-layout

S043: OCR cognitive skill
https://learn.microsoft.com/azure/search/cognitive-search-skill-ocr

Q65 Domain E: Implement information extraction solutions Multiple select

An agent retrieves an unrelated Search index and returns content from another department. Which TWO tool configuration controls address the stated defects? Select TWO answers.

A A resource name containing the department.

B A supported filter using the authorized department. ✓

C A larger unfiltered candidate set.

D The correct connected index name. ✓

E A longer model answer.

Why the correct answer is right

Select the intended index and enforce the supported retrieval filter. The authorized field value comes from the host's access context.

Why the other options are wrong

- **E:** Length does not change retrieved data.
- **A:** A display name does not filter documents.
- **C:** More unfiltered candidates leave the boundary defect.

AI103-E1.05

Sources

S040: Connect an Azure AI Search index to Foundry agents
https://learn.microsoft.com/azure/foundry/agents/how-to/tools/ai-search

S054: Tool best practices for Microsoft Foundry Agent Service
https://learn.microsoft.com/azure/foundry/agents/concepts/tool-best-practice

Q66 Domain E: Implement information extraction solutions Multiple select

A document pipeline must extract invoice fields and support amount verification. Which TWO design choices address those requirements? Select TWO answers.

A Retain source locations for reviewer inspection. ✓

B Treat high confidence as permission to pay.

C Replace field types with filename parsing.

D Define the required field schema. ✓

E Discard coordinates after assigning confidence.

Why the correct answer is right

The schema defines extraction targets; grounding locations connect extracted values to the source. Downstream payment validation is separate.

Why the other options are wrong

- **B:** Confidence is not business authorization.
- **C:** A filename does not supply typed document fields.
- **E:** Discarding locations weakens evidence review.

AI103-E2.01

Sources

S059: Content Understanding document solutions
https://learn.microsoft.com/en-us/azure/ai-services/content-understanding/document/overview

S045: Validate document analyzer quality with confidence, grounding, and labeled samples
https://learn.microsoft.com/azure/ai-services/content-understanding/document/analyzer-improvement

Q67 Domain E: Implement information extraction solutions Multiple select

A RAG pipeline needs a clean document representation before retrieval. Which TWO Content Understanding outputs or behaviors fit? Select TWO answers.

A A completed analysis result consumed after successful polling. ✓

B Markdown representing document structure. ✓

C A generated final answer used as the source document.

D A queued operation treated as extracted text.

E A raw poller serialized into the vector field.

Why the correct answer is right

Structured/Markdown extraction and completed analysis results support downstream processing. An operation handle is not the extracted representation.

Why the other options are wrong

- **D:** Queued work has not yielded a completed representation.
- **C:** A final answer is downstream generation rather than the source representation.
- **E:** The poller is control state, not an embedding vector.

AI103-E2.02

Sources

S059: Content Understanding document solutions
https://learn.microsoft.com/en-us/azure/ai-services/content-understanding/document/overview

S021: Azure AI Content Understanding client library for Python - version 1.1.0
https://learn.microsoft.com/python/api/overview/azure/ai-contentunderstanding-readme?view=azure-python

Q68 Domain E: Implement information extraction solutions Multiple select

A custom contract analyzer misses renewal dates in a new layout. Which TWO inputs should guide improvement? Select TWO answers.

A A typed renewal-date field definition. ✓

B A high threshold reported as proof of correctness.

C Renaming every unrelated date as renewal_date.

D Tune the query semantic configuration without changing the extraction field definition.

E Representative examples with verified values. ✓

Why the correct answer is right

Define the intended field and use representative labeled evidence to improve extraction. Evaluate results on separate documents.

Why the other options are wrong

- **D:** Query ranking cannot define the missing renewal-date extraction target.
- **C:** Renaming an unrelated date does not identify the business field.
- **B:** A confidence threshold is not verification.

AI103-E2.03

Sources

S059: Content Understanding document solutions
https://learn.microsoft.com/en-us/azure/ai-services/content-understanding/document/overview

S045: Validate document analyzer quality with confidence, grounding, and labeled samples
https://learn.microsoft.com/azure/ai-services/content-understanding/document/analyzer-improvement

Q69

Domain B: Implement generative AI and agentic solutions | Multiple select

An agent proposes a refund. The application requires user confirmation, and the model supplies untrusted arguments. Which TWO host responsibilities are needed before the side effect? Select TWO answers.

A Obtain confirmation for the specific proposed refund. ✓

B Return the payment credential to the model for approval checking.

C Execute the refund before displaying its confirmation dialog.

D Validate the arguments and authorize the target account. ✓

E Treat valid JSON as sufficient permission to issue the refund.

Why the correct answer is right

The host must validate and authorize the operation, then obtain confirmation before executing that approved proposal.

Why the other options are wrong

- **E:** Syntax does not establish authorization or approval.
- **C:** Confirmation would occur after the side effect.
- **B:** The host should keep secrets out of tool results.

AI103-B2.01

Sources

S007: Use function calling with Microsoft Foundry agents
https://learn.microsoft.com/azure/foundry/agents/how-to/tools/function-calling

Q70 Domain B: Implement generative AI and agentic solutions Multiple select

A strict Responses output must contain code and comment. Comment may be null, and additional fields are prohibited. Which TWO schema decisions satisfy those constraints? Select TWO answers.

A Allow arbitrary extra fields and validate just the required names.

B Keep comment required and allow its type to include null. ✓

C Represent absent comment as an omitted property with no nullable type.

D Set additionalProperties to false on the object schema. ✓

E Omit comment from required when its value is unavailable.

Why the correct answer is right

Use required nullable fields for optional information and disallow additional properties. The supported strict subset treats presence and nullability separately.

Why the other options are wrong

- **E:** The contract requires the property to be present.
- **A:** The contract prohibits additional fields.
- **C:** That changes both presence and value semantics.

AI103-B3.01

Sources

S037: Structured outputs
https://learn.microsoft.com/azure/foundry/openai/how-to/structured-outputs

Q71

Domain B: Implement generative AI and agentic solutions

Multiple select

A RAG system must find exact equipment identifiers and semantic paraphrases. It uses a vector index built with model X. Which TWO design choices fit? Select TWO answers.

A Use an unrelated model with matching dimensions as proof of vector compatibility.

B Sort hybrid candidates by filename after RRF to preserve relevance.

C Replace embeddings with semantic reranking during ingestion.

D Use query embeddings compatible with the indexed model X vectors. ✓

E Combine keyword and vector retrieval through hybrid search. ✓

Why the correct answer is right

Hybrid retrieval supplies complementary lexical and semantic signals. Vector relevance also depends on compatible query and content representations.

Why the other options are wrong

- **A:** Dimension equality does not establish a shared representation.
- **C:** Reranking does not generate stored vectors.
- **B:** Filename order is not relevance ranking.

AI103-B1.02

Sources

S025: Hybrid search using vectors and full-text search in Azure AI Search
https://learn.microsoft.com/azure/search/hybrid-search-overview

S026: Integrated vector embedding in Azure AI Search
https://learn.microsoft.com/azure/search/vector-search-integrated-vectorization

Q72 Domain B: Implement generative AI and agentic solutions Multiple select

An agent receives hostile user prompts and malicious instructions inside retrieved text. Which TWO controls align with those different entry points? Select TWO answers.

A Treat retrieved-document instructions as lower-trust indirect attacks. ✓

B Approve retrieved instructions because the source URL uses HTTPS.

C Scan final answers as proof that proposed tool calls were safe before execution.

D Treat tool output as a new developer message when it names a system role.

E Assess direct user attacks at the input boundary. ✓

Why the correct answer is right

User input and retrieved content create different attack paths. Apply controls at the relevant boundaries and preserve instruction authority.

Why the other options are wrong

- **D:** Text inside a result does not acquire a privileged role.
- **B:** Transport security does not establish instruction authority.
- **C:** A final scan occurs after earlier tool actions.

AI103-B2.05

Sources

S015: Guardrails and controls overview in Microsoft Foundry
https://learn.microsoft.com/azure/foundry/guardrails/guardrails-overview

S016: Prompt Shields in Microsoft Foundry
https://learn.microsoft.com/azure/foundry/openai/concepts/content-filter-prompt-shields

Q73 Domain B: Implement generative AI and agentic solutions Multiple select

A support agent needs a remembered user preference, current corporate policy and a receipt uploaded for this interaction. Which TWO assignments are appropriate? Select TWO answers.

A Use memory as the maintained full-text corporate policy repository.

B Use the knowledge base for policy and file search for the receipt. ✓

C Use conversation state as the cross-session preference store.

D Use the temporary receipt as the source of the company's standing policy.

E Use persistent memory for interaction-derived preferences. ✓

Why the correct answer is right

Memory, curated knowledge and temporary document search serve different information needs. Assign each source according to its origin and lifecycle.

Why the other options are wrong

- **C:** Session state is short-term context.
- **A:** Curated policy belongs in a maintained knowledge source.
- **D:** The uploaded transaction document is not the curated policy authority.

AI103-B2.02

Sources

S006: Memory in Microsoft Foundry Agent Service (preview)
https://learn.microsoft.com/azure/foundry/agents/concepts/what-is-memory

Q74

Domain B: Implement generative AI and agentic solutions | Multiple select

A proxy calls a memory-enabled prompt agent for two users using one managed identity. Which TWO requirements implement separate preferences? Select TWO answers.

A Use a single static scope for every resident.

B Put user display names only in the prompt.

C Use the memory tool's {{$userId}} scope. ✓

D Rely on new conversation IDs as memory partitions.

E Send a verified user key through x-memory-user-id. ✓

Why the correct answer is right

The tool's scope-resolution mechanism is separate from conversation lifecycle. The backend must provide an authorized end-user key.

Why the other options are wrong

- **A:** A static scope intentionally shares memory.
- **B:** Prompt naming does not select the tool's partition.
- **D:** A conversation ID does not substitute for the memory tool's scope.

AI103-B2.02

Sources

S053: Create and use memory in Foundry Agent Service (preview)
https://learn.microsoft.com/en-us/azure/foundry/agents/how-to/memory-usage

Q75 Domain B: Implement generative AI and agentic solutions Multiple select

A support agent needs a domain boundary and a measurable fallback for missing evidence. Which TWO activities address those requirements? Select TWO answers.

A State the domain and fallback in system instructions. ✓

B Assume one successful example establishes compliance.

C Evaluate representative and adversarial queries against that behavior. ✓

D Remove the fallback to reduce prompt length.

E Treat a low temperature as complete policy enforcement.

Why the correct answer is right

Instructions specify desired behavior, and evaluation checks it. Add host/filter controls when the task requires enforcement.

Why the other options are wrong

- **E:** Temperature does not guarantee policy compliance.
- **B:** One example does not establish behavior across cases.
- **D:** Removing fallback leaves missing-evidence behavior unspecified.

AI103-B3.01

Sources

S057: System message design
https://learn.microsoft.com/en-us/azure/cognitive-services/openai/concepts/advanced-prompt-engineering?pivots=programming-language-chat-completions

Q76 Domain B: Implement generative AI and agentic solutions Multiple select

A specialist agent proposes an action that a deterministic host rule must approve. Which TWO responsibilities should the design keep explicit? Select TWO answers.

A The host’s rule/authorization gate before action execution. ✓

B The orchestration’s handoff and completion conditions. ✓

C The model’s acknowledgement treated as business approval.

D A trace display label treated as an executable rule.

E Tool registration treated as universal caller permission.

Why the correct answer is right

Hybrid orchestration separates proposals from deterministic rules and supplies explicit flow control. Tool availability does not grant permission.

Why the other options are wrong

- **C:** A model statement does not supply the required approval.
- **E:** Registration does not authorize every caller.
- **D:** A label is not a rule implementation.

AI103-B3.04

Sources

S008: Build a workflow in Microsoft Foundry (Preview)
https://learn.microsoft.com/azure/foundry/agents/concepts/workflow

S007: Use function calling with Microsoft Foundry agents
https://learn.microsoft.com/azure/foundry/agents/how-to/tools/function-calling

Q77 Domain D: Implement text analysis solutions Sequence

A Python OpenAI 2.x application extracts events with Responses structured output. The client is configured and the supported model is deployed. On the successful non-refusal path, arrange these operations; business validation needs the parsed field values.

Arrange these steps in order:

A. Read the successful response.output_parsed object.

B. Define the Pydantic event model.

C. Call responses.parse with the event model as text_format.

D. Validate the parsed date and participants against application rules.

Correct order:

1. B Define the Pydantic event model.

2. C Call responses.parse with the event model as text_format.

3. A Read the successful response.output_parsed object.

4. D Validate the parsed date and participants against application rules.

Why the correct answer is right

The type defines the request schema; the response supplies the parsed object. Schema conformance does not validate the event against business rules.

Answer and step notes

- **B:** The event type must exist before it can configure the request.
- **C:** The request consumes that type and returns the response.
- **A:** The successful response supplies the parsed values.
- **D:** Business checks consume those values rather than the request schema.

AI103-D1.01

Sources

S037: Structured outputs
https://learn.microsoft.com/azure/foundry/openai/how-to/structured-outputs

Q78 Domain D: Implement text analysis solutions Sequence

A Language NER pipeline must send organization and location records to a JSON-consuming application. The host constructs the records from returned entity categories and verifies the resulting JSON against the consumer’s schema. Arrange the data dependencies.

Arrange these steps in order:

A. Construct JSON records from the returned organization and location entities.

B. Read the source text into the extraction worker.

C. Run named entity recognition on that text.

D. Validate the constructed JSON against the consumer schema before sending it.

Correct order:

1. B Read the source text into the extraction worker.

2. C Run named entity recognition on that text.

3. A Construct JSON records from the returned organization and location entities.

4. D Validate the constructed JSON against the consumer schema before sending it.

Why the correct answer is right

NER consumes source text and returns categorized entities. The host builds its records from that result, then validates the actual serialized payload for the consumer.

Answer and step notes

- **B:** The text supplies the NER input.
- **C:** NER supplies the categorized entities.
- **A:** The records are constructed from those entities.
- **D:** Validation requires the actual constructed payload.

AI103-D1.01

Sources

S030: What is Azure Language in Foundry Tools?
https://learn.microsoft.com/azure/ai-services/language-service/overview

S037: Structured outputs
https://learn.microsoft.com/azure/foundry/openai/how-to/structured-outputs

Q79 Domain D: Implement text analysis solutions Sequence

A transcript may be held temporarily in a restricted worker. The searchable store may receive only a version redacted using the entities returned by PII detection. The worker has no existing detection result. Arrange the data dependencies.

Arrange these steps in order:

A. Read the transcript into the restricted worker.

B. Build the redacted representation using the returned detected entities.

C. Write that redacted representation to the searchable store.

D. Analyze that text with PII detection.

Correct order:

1. A Read the transcript into the restricted worker.

2. D Analyze that text with PII detection.

3. B Build the redacted representation using the returned detected entities.

4. C Write that redacted representation to the searchable store.

Why the correct answer is right

PII detection identifies entities in the original text. The host uses those results to create the representation required by the stated storage policy before indexing.

Answer and step notes

- **A:** The worker supplies the analysis input.
- **D:** Detection must process the input before its output exists.
- **B:** The detected entities supply the host’s redaction targets.
- **C:** Indexing the original text first would breach the specified boundary.

AI103-D1.02

Sources

S030: What is Azure Language in Foundry Tools?
https://learn.microsoft.com/azure/ai-services/language-service/overview

Q80 Domain D: Implement text analysis solutions Sequence

A mixed-language notice must be translated into French. Its language spans are known, and quality review compares the complete reconstructed notice with the original, including references across segments. Arrange this application workflow.

Arrange these steps in order:

A. Translate each prepared segment to French with its source language specified.

B. Reconstruct the notice from the successful segment translations.

C. Review the complete translated notice against the original before delivery.

D. Prepare single-language segments with their intended source-language labels.

Correct order:

1. D Prepare single-language segments with their intended source-language labels.

2. A Translate each prepared segment to French with its source language specified.

3. B Reconstruct the notice from the successful segment translations.

4. C Review the complete translated notice against the original before delivery.

Why the correct answer is right

Single-language preparation addresses the documented mixture limitation. Reconstruction consumes translations; this review explicitly requires the complete notice.

Answer and step notes

- **D:** Prepared segments and labels are inputs to the translation calls.
- **A:** Translation produces the target-language segments.
- **B:** Reconstruction combines those outputs.
- **C:** The specified cross-segment review needs the reconstructed notice.

AI103-D1.03

Sources

S055: Azure Translator known issues and troubleshooting
https://learn.microsoft.com/en-us/azure/ai-services/translator/reference/known-issues

S032: What is Azure Translator in Foundry Tools?
https://learn.microsoft.com/azure/ai-services/translator/overview

Q81 Domain D: Implement text analysis solutions Sequence

A compliance extractor may forward a result for approval only after its cited fields are verified against the supplied policy. Its model, client and policy are ready, but its domain instructions and strict output schema have not been defined. Arrange these operations.

Arrange these steps in order:

A. Define the domain boundary, missing-evidence fallback and supported field schema.

B. Check the successful extracted fields and citations against the supplied policy.

C. Forward the verified result to the authorized approval queue.

D. Request extraction using those instructions, schema and policy text.

Correct order:

1. A Define the domain boundary, missing-evidence fallback and supported field schema.

2. D Request extraction using those instructions, schema and policy text.

3. B Check the successful extracted fields and citations against the supplied policy.

4. C Forward the verified result to the authorized approval queue.

Why the correct answer is right

Instructions and schema configure extraction. The host verifies evidence before forwarding; neither instructions nor valid JSON replace that gate.

Answer and step notes

- **A:** The request needs the domain and output contracts.
- **D:** Inference produces the candidate extraction.
- **B:** Evidence checks require that candidate and its source.
- **C:** The stated queue gate requires successful verification.

AI103-D1.04

Sources

S057: System message design
https://learn.microsoft.com/en-us/azure/cognitive-services/openai/concepts/advanced-prompt-engineering?pivots=programming-language-chat-completions

S037: Structured outputs
https://learn.microsoft.com/azure/foundry/openai/how-to/structured-outputs

Q82 Domain D: Implement text analysis solutions Sequence

A voice assistant uses a text-only response model and a separate speech synthesizer. The application requires its response-text policy check to pass before any answer audio is emitted. Arrange the successful path for one recorded utterance.

Arrange these steps in order:

A. Generate a text response from that transcript.

B. Recognize the utterance into finalized text.

C. Apply the required response-text policy check.

D. Synthesize and emit the approved response text.

Correct order:

1. B Recognize the utterance into finalized text.

2. A Generate a text response from that transcript.

3. C Apply the required response-text policy check.

4. D Synthesize and emit the approved response text.

Why the correct answer is right

The text-only model needs recognition output. The policy gate consumes its answer and must precede synthesis under this application contract.

Answer and step notes

- **B:** Recognition supplies text to the response model.
- **A:** Generation supplies the candidate answer.
- **C:** The policy check requires the candidate answer.
- **D:** The specified policy allows audio only from approved text.

AI103-D2.01

Sources

S033: What is Azure Speech?
https://learn.microsoft.com/azure/ai-services/speech-service/overview

S057: System message design
https://learn.microsoft.com/en-us/azure/cognitive-services/openai/concepts/advanced-prompt-engineering?pivots=programming-language-chat-completions

Q83 Domain D: Implement text analysis solutions Sequence

A Custom Speech project and training data are ready. A new model must meet a held-out WER threshold before deployment, and the application will use it for real-time recognition, not batch transcription. Arrange the remaining operations.

Arrange these steps in order:

A. Evaluate that model on the held-out set and confirm the WER gate passes.

B. Deploy the approved model to a custom endpoint.

C. Train the custom model with the prepared domain data.

D. Send real-time recognition requests to that custom endpoint.

Correct order:

1. C Train the custom model with the prepared domain data.

2. A Evaluate that model on the held-out set and confirm the WER gate passes.

3. B Deploy the approved model to a custom endpoint.

4. D Send real-time recognition requests to that custom endpoint.

Why the correct answer is right

The model must exist before evaluation. The stated release gate precedes deployment, and real-time use requires a deployed custom endpoint; batch usage has a different requirement.

Answer and step notes

- **C:** Training produces the model being evaluated.
- **A:** Evaluation verifies the specified deployment gate.
- **B:** Deployment creates the endpoint for this model.
- **D:** Real-time requests use that deployed endpoint.

AI103-D2.02

Sources

S034: What is custom speech?
https://learn.microsoft.com/azure/ai-services/speech-service/custom-speech-overview

Q84 Domain D: Implement text analysis solutions Sequence

A Realtime assistant must interpret a recorded inspection question in the context of a written maintenance note for the same equipment. Its supported audio model and transport are ready. Host policy requires checking the returned interpretation against that note before offering a repair action. Arrange the application’s operations.

Arrange these steps in order:

A. Prepare the supported audio input and text context for that equipment’s conversation.

B. Check that interpretation against the maintenance note before offering the repair action.

C. Collect the utterance and its associated maintenance note.

D. Submit the prepared input and obtain the successful interpretation.

Correct order:

1. C Collect the utterance and its associated maintenance note.

2. A Prepare the supported audio input and text context for that equipment’s conversation.

3. D Submit the prepared input and obtain the successful interpretation.

4. B Check that interpretation against the maintenance note before offering the repair action.

Why the correct answer is right

The audio and written context supply the model inputs. The interpretation exists after inference, and the stated host gate consumes it before an action is offered.

Answer and step notes

- **C:** The associated inputs must exist before they can be prepared.
- **A:** Input and conversation context must be prepared before submission.
- **D:** Successful inference supplies the interpretation.
- **B:** The host check consumes that interpretation and its associated note.

AI103-D2.03

Sources

S036: Use the GPT Realtime API for speech and audio
https://learn.microsoft.com/azure/foundry/openai/how-to/realtime-audio

Q85 Domain D: Implement text analysis solutions Sequence

An event-based Speech SDK translator must retain the first completed translated utterance. Credentials and AudioConfig are ready. It uses a TranslationRecognizer whose recognized callback stores successful translations. Arrange initialization so the handler is ready when recognition begins.

Arrange these steps in order:

A. Create the TranslationRecognizer from that translation configuration and AudioConfig.

B. Configure SpeechTranslationConfig with the source and target languages.

C. Connect the storage callback to the recognizer’s recognized event.

D. Start continuous recognition on the configured recognizer.

Correct order:

1. B Configure SpeechTranslationConfig with the source and target languages.

2. A Create the TranslationRecognizer from that translation configuration and AudioConfig.

3. C Connect the storage callback to the recognizer’s recognized event.

4. D Start continuous recognition on the configured recognizer.

Why the correct answer is right

The configuration is consumed by the constructor. The event belongs to the created recognizer, and this application must subscribe before starting to avoid losing its first result.

Answer and step notes

- **B:** The constructor needs the completed translation configuration.
- **A:** The recognizer must exist before its event can be subscribed.
- **C:** The application requires the callback to be ready before recognition starts.
- **D:** Recognition now delivers completed results to the installed handler.

AI103-D2.04

Sources

S046: How to recognize and translate speech
https://learn.microsoft.com/azure/ai-services/speech-service/how-to-translate-speech

Q86 Domain D: Implement text analysis solutions Sequence

A store-and-forward speech translator uses manual synthesis for two target languages. It will deliver an audio file only after synthesis succeeds. Arrange the successful path for the selected target language; no translation result is available initially.

Arrange these steps in order:

A. Synthesize that text with a voice supporting the selected target language.

B. Deliver the successfully synthesized audio file.

C. Select the requested language's text from the successful result's translations.

D. Recognize and translate the source utterance.

Correct order:

1. D Recognize and translate the source utterance.

2. C Select the requested language's text from the successful result's translations.

3. A Synthesize that text with a voice supporting the selected target language.

4. B Deliver the successfully synthesized audio file.

Why the correct answer is right

The translation result supplies per-language text. Manual synthesis consumes the selected text; this file-delivery contract requires completed audio rather than a streamed partial response.

Answer and step notes

- **D:** Translation produces the per-language results.
- **C:** Selection supplies the requested target text.
- **A:** Synthesis consumes that text and produces audio.
- **B:** The application delivers a file only after successful synthesis.

AI103-D2.04

Sources

S046: How to recognize and translate speech
https://learn.microsoft.com/azure/ai-services/speech-service/how-to-translate-speech

S033: What is Azure Speech?
https://learn.microsoft.com/azure/ai-services/speech-service/overview

Q87 Domain A: Plan and manage an Azure AI solution Matching

Match each workload requirement to its deployment family. Geography and model support have already been checked. Use each selected choice once; one choice is unused.

Match each row to a choice.

A. Deferred asynchronous groups of inference requests
Match: _____

B. Variable interactive demand with per-token charges
Match: _____

C. Allocated throughput for sustained traffic
Match: _____

Choice bank

- Developer
- Batch
- Provisioned
- Standard

Correct matches:

A. Deferred asynchronous groups of inference requests → Batch

B. Variable interactive demand with per-token charges → Standard

C. Allocated throughput for sustained traffic → Provisioned

Why the correct answer is right

Standard is usage-based interactive serving; Provisioned allocates throughput; Batch serves asynchronous work. Developer is the unused experimentation choice.

Answer and step notes

- **B:** Variable interactive demand with per-token charges: Standard.
- **C:** Allocated throughput for sustained traffic: Provisioned.
- **A:** Deferred asynchronous groups of inference requests: Batch.

AI103-A2.02

Sources

S005: Deployment types for Microsoft Foundry Models
https://learn.microsoft.com/azure/foundry/foundry-models/concepts/deployment-types

Q88 Domain A: Plan and manage an Azure AI solution Matching

Match each investigation to the most direct evidence source. Use each selected choice once; one choice is unused.

Match each row to a choice.

A. A slow operation inside one scored agent request
Match: _____

B. Missing source documents after a partially successful run
Match: _____

C. Query-serving throttling during traffic peaks
Match: _____

Choice bank

- Search service operational metrics
- Indexer item errors and warnings
- Correlated trace and nested spans
- Aggregate billing report

Correct matches:

A. A slow operation inside one scored agent request	**→ Correlated trace and nested spans**
B. Missing source documents after a partially successful run	**→ Indexer item errors and warnings**
C. Query-serving throttling during traffic peaks	**→ Search service operational metrics**

Why the correct answer is right

Use operation traces for request-level timing, indexer details for ingestion failures, and service metrics for serving health. An aggregate billing report cannot localize these request, ingestion or serving failures.

Answer and step notes

- **A:** A slow operation inside one scored agent request: Correlated trace and nested spans.
- **B:** Missing source documents after a partially successful run: Indexer item errors and warnings.
- **C:** Query-serving throttling during traffic peaks: Search service operational metrics.

AI103-A3.03

Sources

S013: Agent tracing overview
https://learn.microsoft.com/azure/foundry/observability/concepts/trace-agent-concept

S027: Monitor Azure AI Search
https://learn.microsoft.com/azure/search/monitor-azure-cognitive-search

S044: Monitor indexer status and results in Azure AI Search
https://learn.microsoft.com/azure/search/search-monitor-indexers

Q89 Domain A: Plan and manage an Azure AI solution Matching

Match each keyless inference observation to the boundary to inspect. Use each selected choice once; one choice is unused.

Match each row to a choice.

A. A token is acquired but the resource denies inference
Match: _____

B. The role exists only on a different resource
Match: _____

C. No Entra token can be acquired
Match: _____

Choice bank

- Credential/authentication configuration
- Inference authorization for the caller
- Role assignment scope covering the called resource
- Generation sampling parameters

Correct matches:

A. A token is acquired but the resource denies inference	**→ Inference authorization for the caller**
B. The role exists only on a different resource	**→ Role assignment scope covering the called resource**
C. No Entra token can be acquired	**→ Credential/authentication configuration**

Why the correct answer is right

Authentication, permission and scope are separate boundaries.

Answer and step notes

- **C:** No Entra token can be acquired: Credential/authentication configuration.
- **A:** A token is acquired but the resource denies inference: Inference authorization for the caller.
- **B:** The role exists only on a different resource: Role assignment scope covering the called resource.

AI103-A3.04

Sources

S051: Azure OpenAI v1 API
https://learn.microsoft.com/en-us/azure/foundry/openai/api-version-lifecycle

S052: Use Azure OpenAI without keys
https://learn.microsoft.com/en-us/azure/developer/ai/keyless-connections

S010: Role-based access control for Microsoft Foundry
https://learn.microsoft.com/azure/foundry/concepts/rbac-foundry

Q90 Domain A: Plan and manage an Azure AI solution Matching

Match each trust boundary to the relevant inspection point. Use each selected choice once; one choice is unused.

Match each row to a choice.

A. An instruction inserted into a retrieved document
Match: _____

B. A harmful generated response
Match: _____

C. A user’s submitted attack instruction
Match: _____

Choice bank

- User-input protection
- External-context/indirect-attack protection
- Final-output inspection
- Model token quota

Correct matches:

A. An instruction inserted into a retrieved document	**→ External-context/indirect-attack protection**
B. A harmful generated response	**→ Final-output inspection**
C. A user’s submitted attack instruction	**→ User-input protection**

Why the correct answer is right

Protect each content boundary; a safe earlier input does not establish safety of later content.

Answer and step notes

- **C:** A user’s submitted attack instruction: User-input protection.
- **A:** An instruction inserted into a retrieved document: External-context/indirect-attack protection.
- **B:** A harmful generated response: Final-output inspection.

AI103-A4.01

Sources

S015: Guardrails and controls overview in Microsoft Foundry
https://learn.microsoft.com/azure/foundry/guardrails/guardrails-overview

S016: Prompt Shields in Microsoft Foundry
https://learn.microsoft.com/azure/foundry/openai/concepts/content-filter-prompt-shields

Q91 Domain B: Implement generative AI and agentic solutions Matching

Match each observed agent defect to the targeted tool evaluator. Use each selected choice once; one choice is unused.

Match each row to a choice.

A. A correct returned quantity is misrepresented in the answer
Match: _____

B. The model invokes translation when inventory lookup is needed
Match: _____

C. The correct tool receives the warehouse ID as a part ID
Match: _____

Choice bank

- Tool Input Accuracy
- Task Completion (preview)
- Tool Output Utilization
- Tool Selection

Correct matches:

A. A correct returned quantity is misrepresented in the answer → Tool Output Utilization

B. The model invokes translation when inventory lookup is needed → Tool Selection

C. The correct tool receives the warehouse ID as a part ID → Tool Input Accuracy

Why the correct answer is right

Selection concerns the chosen capability, input accuracy concerns arguments, and output utilization concerns returned information. Task Completion is a broader outcome metric.

Answer and step notes

- **B:** The model invokes translation when inventory lookup is needed: Tool Selection.
- **C:** The correct tool receives the warehouse ID as a part ID: Tool Input Accuracy.
- **A:** A correct returned quantity is misrepresented in the answer: Tool Output Utilization.

AI103-B2.06

Sources

S014: Agent evaluators
https://learn.microsoft.com/azure/foundry/concepts/evaluation-evaluators/agent-evaluators

Q92 Domain B: Implement generative AI and agentic solutions Matching

Match each information need to the documented context capability. Use each selected choice once; one choice is unused.

Match each row to a choice.

A. A maintained corporate handbook
Match: _____

B. A preference learned from prior conversations
Match: _____

C. A receipt uploaded for the current interaction
Match: _____

Choice bank

- Persistent memory
- Model router
- Curated knowledge base
- File search

Correct matches:

A. A maintained corporate handbook	**→ Curated knowledge base**
B. A preference learned from prior conversations	**→ Persistent memory**
C. A receipt uploaded for the current interaction	**→ File search**

Why the correct answer is right

Memory, curated knowledge and file search serve different origins and lifecycles of information. Routing selects among models.

Answer and step notes

- **B:** A preference learned from prior conversations: Persistent memory.
- **A:** A maintained corporate handbook: Curated knowledge base.
- **C:** A receipt uploaded for the current interaction: File search.

AI103-B2.02

Sources

S006: Memory in Microsoft Foundry Agent Service (preview)
https://learn.microsoft.com/azure/foundry/agents/concepts/what-is-memory

Q93 Domain B: Implement generative AI and agentic solutions Matching

Match each response-turn requirement to the documented tool_choice mode. Use each selected choice once; one choice is unused.

Match each row to a choice.

A. The turn must produce no tool calls
Match: _____

B. One or more tool calls are necessary
Match: _____

C. Model may decide whether a tool helps
Match: _____

Choice bank

- auto
- required
- none
- A specific tool execution receipt

Correct matches:

A. The turn must produce no tool calls	**→ none**
B. One or more tool calls are necessary	**→ required**
C. Model may decide whether a tool helps	**→ auto**

Why the correct answer is right

Request modes control tool invocation intent; execution and receipts remain host responsibilities.

Answer and step notes

- **C:** Model may decide whether a tool helps: auto.
- **B:** One or more tool calls are necessary: required.
- **A:** The turn must produce no tool calls: none.

AI103-B2.01

Sources

S054: Tool best practices for Microsoft Foundry Agent Service
https://learn.microsoft.com/azure/foundry/agents/concepts/tool-best-practice

Q94 Domain C: Implement computer vision solutions Matching

Match each webpage image role to the appropriate alternative-text approach. Use each selected choice once; one choice is unused.

Match each row to a choice.

A. A complex chart whose details cannot fit concise alt text
Match: _____

B. An image that submits a search form
Match: _____

C. A purely decorative background flourish
Match: _____

Choice bank

- Provide the detailed information elsewhere on the page
- Use the image filename
- Communicate the action
- Use empty alt

Correct matches:

A. A complex chart whose details cannot fit concise alt text	**→ Provide the detailed information elsewhere on the page**
B. An image that submits a search form	**→ Communicate the action**
C. A purely decorative background flourish	**→ Use empty alt**

Why the correct answer is right

Image purpose determines its alternative. A complex chart needs access to its detailed information in addition to a suitable concise alternative.

Answer and step notes

- **B:** An image that submits a search form: Communicate the action.
- **C:** A purely decorative background flourish: Use empty alt.
- **A:** A complex chart whose details cannot fit concise alt text: Provide the detailed information elsewhere on the page.

AI103-C2.04

Sources

S050: W3C WAI An alt Decision Tree
https://www.w3.org/WAI/tutorials/images/decision-tree/

Q95 Domain E: Implement information extraction solutions Configuration

The index has no OCR text. Which one configuration defect in this exhibit blocks the OCR input path?

Blob OCR pipeline

```
{
  "indexer": {
    "parameters": {"configuration": {"imageAction": "none"}},
    "outputFieldMappings": [{
      "sourceFieldName": "/document/normalized_images/*/text",
      "targetFieldName": "ocr_text"
    }]
  },
  "skill": {
    "@odata.type": "#Microsoft.Skills.Vision.OcrSkill",
    "context": "/document/normalized_images/*",
    "inputs": [{"name": "image", "source": "/document/normalized_images/*"}],
    "outputs": [{"name": "text", "targetName": "text"}]
  },
  "indexField": {"name": "ocr_text", "type": "Collection(Edm.String)", "searchable": true}
}
```

A The index’s ocr_text field is searchable.

B The text output mapping names the correct enriched node.

C The blob indexer’s imageAction is none. ✓

D The OCR skill uses the normalized image wildcard.

Why the correct answer is right

The normalized image collection must exist for the OCR skill input. The exhibit’s output path is already correct.

Why the other options are wrong

- **B:** This correctly maps the produced node after the skill runs.
- **A:** Searchability is appropriate for retrieval.
- **D:** The wildcard matches the skill’s normalized-image input.

AI103-E1.04

Sources

S043: OCR cognitive skill
https://learn.microsoft.com/azure/search/cognitive-search-skill-ocr

S028: Skillset concepts in Azure AI Search
https://learn.microsoft.com/azure/search/cognitive-search-working-with-skillsets

Q96 Domain E: Implement information extraction solutions Configuration

The pipeline needs layout-derived Markdown for indexing. Which producing capability should replace the enrichment stage in this exhibit?

Required enrichment output


```
{"input":"source document","requiredOutput":"layout-aware Markdown","consumer":"Search indexer skillset"}
```


A A field mapper without a producing skill.

B A semantic query configuration.

C OCR only, with no layout extraction.

D Document Intelligence Layout skill. ✓

Why the correct answer is right

The Layout skill extracts document layout into Markdown and related structure for enrichment. OCR alone does not supply this layout-aware representation.

Why the other options are wrong

- **C:** OCR supplies text, not the requested layout capability.
- **B:** Query-time semantics do not produce ingestion output.
- **A:** A mapping cannot create the missing representation.

AI103-E1.03

Sources

S029: Document Layout skill
https://learn.microsoft.com/azure/search/cognitive-search-skill-document-intelligence-layout

S059: Content Understanding document solutions
https://learn.microsoft.com/en-us/azure/ai-services/content-understanding/document/overview

Q97 Domain E: Implement information extraction solutions Configuration

A custom extractor uses the following result. What should happen before the application posts the invoice amount?

Invoice analyzer result


```
{"total":{"value":1460.00,"confidence":0.42,"source":{"page":2,"span":"invoice total"}},"rule":"verifie
d totals only"}
```


A Verify grounded field evidence and business rules. ✓

B Use the confidence as payment authorization.

C Replace the source amount with the highest nearby number.

D Post because the JSON parsed.

Why the correct answer is right

The result carries low confidence and source grounding for inspection. Extraction does not authorize payment or resolve the business rule.

Why the other options are wrong

- **D:** Parsing verifies shape, not the amount or payment permission.
- **B:** Confidence estimates extraction, not authority.
- **C:** A nearby number need not be the required field.

AI103-E2.03

Sources

S045: Validate document analyzer quality with confidence, grounding, and labeled samples
https://learn.microsoft.com/azure/ai-services/content-understanding/document/analyzer-improvement

S059: Content Understanding document solutions
https://learn.microsoft.com/en-us/azure/ai-services/content-understanding/document/overview

Q98 Domain C: Implement computer vision solutions Configuration

The image:analyze 2024-09-01 contract uses FourSeverityLevels. Policy rejects an image when any harm category is Medium or High. What comparison should implement the policy?

Image harm policy

```
{"apiVersion":"2024-09-01","outputType":"FourSeverityLevels","rejectWhen":"any category is Medium or High"}
```

A Reject if any analyzed category has severity 4 or 6. ✓

B Average the returned category severities and reject when that average reaches 4.

C Reject when every category equals 6.

D Reject when a category equals 3.

Why the correct answer is right

The image API's severity values are 0, 2, 4 and 6. The stated any-category policy rejects Medium (4) and High (6).

Why the other options are wrong

- **D:** The documented four-level values are 0, 2, 4 and 6.
- **B:** Averages can conceal one category that independently meets the rejection condition.
- **C:** The policy triggers on any Medium or High category.

AI103-C3.01

Sources

S042: QuickStart: Analyze image content
https://learn.microsoft.com/azure/ai-services/content-safety/quickstart-image

Q99 Domain C: Implement computer vision solutions Configuration

This audio-analysis worker passes an operation handle to its transcript consumer and the consumer rejects it. Which change obtains the completed analysis output in the pinned SDK?

Python SDK 1.1.0 / API 2025-11-01

```
poller = client.begin_analyze(analyzer_id="prebuilt-audioSearch", inputs=inputs)
transcript_consumer(poller)
```

A Wait for poller.result() and consume the returned analysis result. ✓

B Read the operation ID as the transcript and retry the consumer.

C Cast the poller to a dictionary before sending it to the consumer.

D Change the audio filename extension and reuse the operation handle.

Why the correct answer is right

begin_analyze returns a long-running operation poller. The SDK example waits with result() before using completed content; the operation handle is not a transcript.

Why the other options are wrong

- **B:** The operation ID identifies work, not the analyzed content.
- **C:** Casting a control object does not obtain successful analysis output.
- **D:** Renaming the input does not complete or retrieve the existing operation.

AI103-C2.05

Sources

S021: Azure AI Content Understanding client library for Python - version 1.1.0
https://learn.microsoft.com/python/api/overview/azure/ai-contentunderstanding-readme?view=azure-python

Q100 Domain D: Implement text analysis solutions Configuration

The translation result leaves part of this input unchanged. Which cause and response best fit the documented condition?

Text translation diagnosis


```
{"text":"Replace the filter, dann starten Sie die Pumpe.","detectedLanguage":"en","target":"fr","sympto
m":"German clause unchanged"}
```


A A detected language and HTTP success establish complete translation.

B Source autodetection must be forced to German for the whole sentence.

C A text request should become an autodetected document translation job.

D Mixed source languages; segment and translate with intended source labels. ✓

Why the correct answer is right

The known-issues page identifies mixed-language sentences as a text translation limitation. A detected language for the whole input does not resolve the mixture.

Why the other options are wrong

- **B:** Choosing one language for the whole mixture does not separate its English span.
- **C:** Changing endpoints while retaining the mixture does not resolve the documented condition.
- **A:** Detection and transport success do not verify every span was translated.

AI103-D1.03

Sources

S055: Azure Translator known issues and troubleshooting
https://learn.microsoft.com/en-us/azure/ai-services/translator/reference/known-issues

Quick Answer Key

#	Type	Correct Answer
Q1	Case study	**C**
Q2	Case study	**B**
Q3	Case study	**B**
Q4	Case study	**A**
Q5	Case study	**B**
Q6	Case study	**C**
Q7	Case study	**A**
Q8	Case study	**D**
Q9	Case study	**B**
Q10	Case study	**C**
Q11	Case study	**D**
Q12	Case study	**C**
Q13	Case study	**B**
Q14	Case study	**B**
Q15	Case study	**D**
Q16	Case study	**C**
Q17	Single choice	**A**
Q18	Single choice	**D**
Q19	Single choice	**D**
Q20	Single choice	**A**
Q21	Single choice	**C**
Q22	Single choice	**A**
Q23	Single choice	**C**
Q24	Single choice	**C**
Q25	Single choice	**D**
Q26	Single choice	**C**
Q27	Configuration	**operation → get | client → get_openai_client**
Q28	Configuration	**scope → {{$userId}} | header → x-memory-user-id**

#	Type	Correct Answer
Q29	Configuration	**mode → required**
Q30	Configuration	**required → ["note"] | extra → False | location → format**
Q31	Single choice	**B**
Q32	Single choice	**A**
Q33	Single choice	**C**
Q34	Single choice	**C**
Q35	Single choice	**C**
Q36	Single choice	**A**
Q37	Single choice	**C**
Q38	Single choice	**A**
Q39	Single choice	**D**
Q40	Single choice	**D**
Q41	Single choice	**C**
Q42	Single choice	**D**
Q43	Configuration	**audience → https://ai.azure.com/.default | endpoint → https://resource.openai.azure.com/openai/v1/**
Q44	Configuration	**endpoint → project_endpoint | credential → DefaultAzureCredential()**
Q45	Single choice	**C**
Q46	Single choice	**A**
Q47	Single choice	**D**
Q48	Single choice	**C**
Q49	Single choice	**A**
Q50	Single choice	**D**
Q51	Single choice	**C**
Q52	Single choice	**A**
Q53	Single choice	**B**
Q54	Single choice	**B**
Q55	Configuration	**kind → image_url | field → image_url**
Q56	Configuration	**source → source_image_file | mask → matching_mask_file**
Q57	Single choice	**D**

#	Type	Correct Answer
Q58	Single choice	**B**
Q59	Multiple select	**E, B**
Q60	Multiple select	**A, C**
Q61	Multiple select	**B, E**
Q62	Multiple select	**A, E**
Q63	Multiple select	**A, D**
Q64	Multiple select	**B, A**
Q65	Multiple select	**D, B**
Q66	Multiple select	**D, A**
Q67	Multiple select	**B, A**
Q68	Multiple select	**A, E**
Q69	Multiple select	**D, A**
Q70	Multiple select	**B, D**
Q71	Multiple select	**E, D**
Q72	Multiple select	**E, A**
Q73	Multiple select	**E, B**
Q74	Multiple select	**C, E**
Q75	Multiple select	**A, C**
Q76	Multiple select	**A, B**
Q77	Sequence	**1. B → 2. C → 3. A → 4. D**
Q78	Sequence	**1. B → 2. C → 3. A → 4. D**
Q79	Sequence	**1. A → 2. D → 3. B → 4. C**
Q80	Sequence	**1. D → 2. A → 3. B → 4. C**
Q81	Sequence	**1. A → 2. D → 3. B → 4. C**
Q82	Sequence	**1. B → 2. A → 3. C → 4. D**
Q83	Sequence	**1. C → 2. A → 3. B → 4. D**
Q84	Sequence	**1. C → 2. A → 3. D → 4. B**
Q85	Sequence	**1. B → 2. A → 3. C → 4. D**
Q86	Sequence	**1. D → 2. C → 3. A → 4. B**
Q87	Matching	**B → Standard | C → Provisioned | A → Batch**

#	Type	Correct Answer
Q88	Matching	**A → Correlated trace and nested spans | B → Indexer item errors and warnings | C → Search service operational metrics**
Q89	Matching	**C → Credential/authentication configuration | A → Inference authorization for the caller | B → Role assignment scope covering the called resource**
Q90	Matching	**C → User-input protection | A → External-context/indirect-attack protection | B → Final-output inspection**
Q91	Matching	**B → Tool Selection | C → Tool Input Accuracy | A → Tool Output Utilization**
Q92	Matching	**B → Persistent memory | A → Curated knowledge base | C → File search**
Q93	Matching	**C → auto | B → required | A → none**
Q94	Matching	**B → Communicate the action | C → Use empty alt | A → Provide the detailed information elsewhere on the page**
Q95	Configuration	**C**
Q96	Configuration	**D**
Q97	Configuration	**A**
Q98	Configuration	**A**
Q99	Configuration	**A**
Q100	Configuration	**D**